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SegFormer with Hausdorff distance-based spatial bias and conceptual split gate for automated segmentation of cervical precancerous lesions

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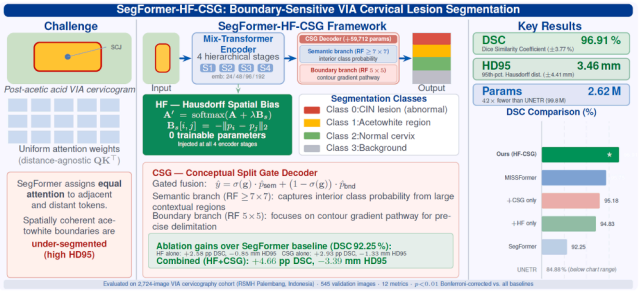
GRAPHICAL ABSTRACT



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ABSTRACT

Automated segmentation of acetowhite lesions in Visual Inspection with Acetic Acid (VIA) cervicography is critical for colposcopy-guided biopsy targeting in low-resource cervical cancer screening programs. Standard transformer architectures compute attention solely from feature content, neglecting spatial proximity between tokens, while unified decoder designs conflate semantic and boundary channels—leaving boundary accuracy at the squamocolumnar junction unresolved.

We propose **SegFormer-HF-CSG**, a parameter-efficient framework that addresses these limitations through two targeted modifications to the SegFormer backbone (as Illustrated in Figure 1):

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Acetowhite lesion segmentation
Squamocolumnar junction delineation
Mix-transformer encoder
Boundary-sensitive decoder
Surface distance metric optimization
Cervicography
Medical image segmentation

- **Hausdorff Distance-based Spatial Bias (HF):** injects pairwise spatial proximity information into the self-attention mechanism across all encoder stages, improving geometric coherence of acetowhite boundary predictions without adding trainable parameters.
- **Conceptual Split Gate (CSG):** decouples the decoder into independent semantic and boundary pathways with adaptive soft gating, adding only 59,712 parameters (+2.3%).

Evaluated on 545 VIA cervicography images, SegFormer-HF-CSG achieves DSC of 96.91%, HD95 of 3.46 mm, and Normalized Surface Distance of 98.82%, outperforming eight reference architectures spanning convolutional, hybrid, and transformer families ($p < 0.01$, Bonferroni-corrected) at a total of 2.62 M parameters—the smallest footprint in the evaluated set.

Specifications table

Subject area	Engineering
More specific subject area	*Pixel-level semantic segmentation of acetowhite lesion regions from Visual Inspection with Acetic Acid (VIA) cervicography images, specifically for automated detection of cervical precancerous lesions (CIN grades) in low-resource clinical screening settings. This is within the domain of parameter-efficient transformer-based medical image segmentation.*
Name of your method	*SegFormer-HF-CSG — SegFormer enhanced with two novel components:* *HF = Hausdorff Distance-Based Spatial Bias (distance-weighted attention bias injected at each Mix-Transformer encoder stage)* *CSG = Conceptual Split Gate (boundary-aware gating in the MLP decoder head)*
Name and reference of original method	*Siti Nurmaini, Patiyus Agustiyansyah, Muhammad Naufal Rachmatullah:"Robust assessment of cervical precancerous lesions from pre- and post-acetic acid cervicography by combining deep learning and medical guidelines", 2025 Informatics in Medicine Unlocked, vol. 52, Issue June 2024, p.101609* [1] *E. Xie, W. Wang, Z. Yu, A. Anandkumar, J. M. Alvarez, and P. Luo, "SegFormer: Simple and efficient design for semantic segmentation with transformers," Advances in Neural Information Processing Systems, vol. 34, pp. 12,077–12,090, 2021* [2].
Resource availability	*Dataset: VIA cervicography corpus from Dr. Mohammad Hoesin General Hospital (RSMH), Palembang, Indonesia — 261 abnormal cases (522 paired pre/post-VIA images), expanded to 2724; ethics approval No DP.04.03/D.XVIII.6.1.1/E-TIK/95/2023.* *Annotation tool: Roboflow (pixel-level polygon annotation). Deep learning framework: TensorFlow 2.13.0 + Keras API, Python 3.8* *Hardware and Software: All experiments were conducted on a computational system featuring an Intel(R) Core (TM) i5–14400F processor, a B760M K V2 DDR4 motherboard, and running on Windows 11.*

Background

Automated segmentation of acetowhite lesion regions in visual inspection with acetic acid (VIA) cervicography plays a vital role in cervical cancer control programs, with direct applications in colposcopy-guided biopsy targeting, posttreatment margin surveillance, and automated triage in low-resource screening settings [1]. Cervical cancer ranked as the fourth most common malignancy among women in 2022, with 662,301 newly diagnosed cases and 348,874 deaths documented worldwide [3]; it constitutes a leading cause of cancer mortality in lower-resource settings, where age-standardized incidence rates reach 23.8 per 100,000 in low Human Development Index (HDI) countries—approximately 2.6 times the 9.3 per 100,000 observed in very-high-HDI nations-and mortality rates are nearly five times higher, reflecting the well-established inverse correlation between socioeconomic development and cervical cancer burden [1]. Of the 185 countries monitored by the World Health Organization (WHO) Cervical Cancer Elimination Initiative, 174 still surpass the four-cases-per- 100,000 elimination threshold; Papanicolaou cytology infrastructure remains inaccessible across much of sub-Saharan Africa and South Central Asia, and VIA-requiring only 3–5% acetic acid and a portable light source-thus retains its role as the WHO-endorsed frontline screening modality for resource-constrained cervical cancer control [3]. Automated VIA segmentation is thus directly linked to public-health impact at population scale. Typically, employing an encoder-decoder structure, current segmentation pipelines decompose the task into hierarchical convolutional feature extraction and up sampled pixel-wise label reconstruction connected by skip pathways. Early methods based on fully convolutional architectures [4] applied successive spatial pooling to extend the receptive field and used transposed convolutions to recover spatial resolution. However, due to the spatially constrained nature of local convolution kernels, their capacity to model coherent boundary relationships across the spatially extended acetowhite transformation zone was limited, thus restricting delineation accuracy at the squamocolumnar junction. To improve contextual modelling, researchers proposed various approaches, notably including channel-wise attention gating and multi-scale pyramid pooling. Nevertheless, these methods have their respective limitations [5,6]:

- Convolutional architectures, while achieving competitive overlap metrics on well-contrasted lesions, suffer from receptive fields that cannot simultaneously model the complete acetowhite transformation zone boundary, progressive resolution loss in the down sampling path that eliminates fine-grained edge gradients, and geometric rigidity that prevents distance weighted encoding of spatial token proximity.
- Attention-augmented convolutional networks, although capable of selectively weighting feature contributions, lead to quadratic complexity at high-resolution encoder stages, residual confounding between boundary-gradient channels and interior semantic

channels in the shared decoder representation, and insufficient boundary-specific gradient supervision when a single cross-entropy (CE) objective drives the full decoder.

In contrast, transformer-based architectures [7] have demonstrated outstanding capabilities in global context modelling, particularly in capturing long-range token dependencies and successfully extending to medical image segmentation [5,6,8]. However, empirical evidence suggests that self-attention scoring is primarily limited to query-key feature similarity, neglecting spatial proximity as an independent weighting factor, overlooking the geometric coherence of post-acetic-acid acetowhite regions (i.e., treating spatially contiguous and spatially distant tokens identically when feature embeddings coincide). To address this, Nurmaini et al. [1] proposed a deep learning framework that incorporates clinical-guideline pre-processing priors for VIA cervicography, which mitigates the domain-adaptation gap caused by acetic acid response variability across patients. Building on this, we introduce the Hausdorff Distance-based Spatial Bias to capture spatial proximity relationships between acetowhite lesion tokens, geometric boundary coherence along the transformation zone perimeter, and post-VIA texture gradient continuity simultaneously.

Recent research has explored the design of hybrid and attention-enriched architectures to combine the advantages of local convolutional inductive bias and global transformer context for medical image segmentation [9–11]. Among these, ViTRNet [10] utilizes a dual-encoder combining RepVGG and Vision Transformer with a Triple Fusion Strategy for multimodal magnetic resonance imaging (MRI) segmentation, while VSmTrans [9] introduces Variable-Shape Mixed Transformer blocks to balance global and local representations in 3D volumetric settings. More recently, ECM-TransUNet [12] integrated edge enhanced multi-scale attention with convolutional Mamba blocks to address boundary ambiguity in computed tomography (CT) and MRI, and FE-MAANet [13] combined multi-scale adaptive large-kernel convolutions with frequency-domain attention for flexible object-scale handling (Fig. 1).

These approaches have significantly improved segmentation quality through architectural specialization and boundary-sensitive training strategies. It is noteworthy that although these methods have demonstrated competitive performance on standard volumetric benchmarks, there remains room for refinement when addressing the unique spatial topology requirements of two-dimensional VIA cervicography with four-class label imbalance (≈ 64% background pixels). Specifically, transformer segmentation architectures in the medical domain (e.g., TransUNet [5] and SwinUNet [8]) apply positional-invariant attention scoring, which fails to encode the spatial coherence of the acetowhite transformation zone a limitation that leaves 95th-percentile boundary displacement (HD95) unaddressed by feature-level architectural modification alone. While global attention effectively captures semantic correspondence, distance-agnostic weighting may compromise HD95 precision when processing spatially extended lesion regions with low texture contrast against adjacent squamous epithelium (e.g., post-VIA Class 1 vs. Class 0 boundaries). Furthermore, the prevalent unified decoder representations employed in hybrid architectures still require further refinement in semantic-boundary signal decomposition (such as separating feature gradients encoding interior class probability from those encoding contour direction), which proves particularly critical for achieving sub-millimeter boundary accuracy in post-VIA acetowhite delineation.

In this paper, we propose SegFormer-HF-CSG, a parameter-efficient cervical lesion segmentation framework that integrates a

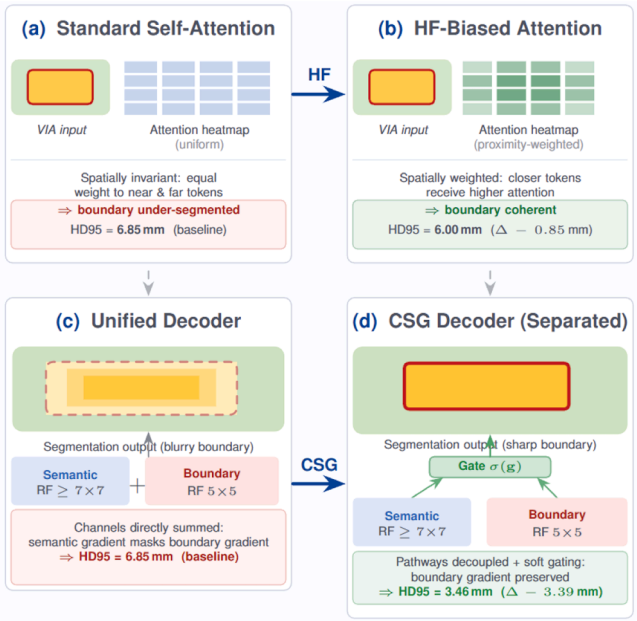

Fig. 1. Illustration of the core motivation for SegFormer-HF-CSG. (a) Standard self-attention produces spatially uniform attention heatmaps, resulting in under-segmented boundaries (HD95 = 6.85 mm). (b) HF-biased attention concentrates proximity-weighted responses at the transformation zone (HD95 = 6.00 mm). (c) Unified decoder conflates semantic and boundary channels (blurry boundary). (d) CSG decoder with decoupled pathways and soft gating preserves boundary gradients (HD95 = 3.46 mm).

Hausdorff Distance-based Spatial Bias (HF) and a Conceptual Split Gate (CSG) into the SegFormer backbone for VIA cervicography processing as illustrated in First, the HF module injects proximity-weighted geometric distance priors directly into the self-attention score matrix at each encoder stage, encoding spatial coherence of the transformation zone without introducing any trainable parameters. Subsequently, within the decoder, the CSG module decomposes the unified feature representation into independent semantic and boundary branches through separate gradient paths, adding only 59,712 parameters (+ 2.3%). Evaluation on a 2724 -image VIA cervicography cohort from Dr. Mohammad Hoesin General Hospital, Palembang-with 545 images reserved for validation-across twelve segmentation metrics consistently demonstrates its superiority over eight reference architectures spanning convolutional, hybrid, and pure-transformer design families.

The contributions of this paper are as follows:

- The Hausdorff Distance-based Spatial Bias (HF) is introduced into the SegFormer encoder, which captures spatial proximity of lesion tokens, boundary coherence along the acetowhite transformation zone, and post-acetic-acid texture continuity, alleviating the limitations of positional-invariant self-attention in standard transformer segmentation.
- A Conceptual Split Gate (CSG) decoder module is designed to address the issue of semantic-boundary signal conflation in the unified feature representations of conventional transformer decoder architectures, enabling independent gradient supervision of interior and contour-sensitive feature channels.
- To address the absence of domain-adapted geometric proximity priors in lightweight transformer segmentation applied to 2D VIA cervicography, we propose the SegFormer-HF-CSG framework trained under a single cross-entropy objective without boundary-weighted auxiliary terms, demonstrating that structural inductive bias alone suffices to outperform architectures with 42 × more parameters on HD95.
- In the experimental evaluation, a twelve-metric protocol spanning region overlap (DSC, IoU, BIoU), boundary distance (HD95, ASD, NSD), and pixel-level classification (Accuracy, Sensitivity, Precision, Specificity, NPV, F1) is applied to a 2724-image single-institution VIA dataset with four-variant controlled ablation, enabling mechanistic attribution of HF and CSG contributions independently.

CNN-Based and transformer-based medical image segmentation

Due to their strong local feature extraction capabilities, convolutional neural networks (CNNs) have been widely adopted in medical image segmentation tasks, establishing the encoder-decoder paradigm as the dominant architectural template in the field. The U-Net framework [4] introduced symmetric skip connections that propagate high-resolution encoder feature maps directly to decoder layers, enabling precise boundary recovery despite progressive down sampling. Building on this, Bi et al. [14] proposed BPAT-UNet, an assembled transformer UNet that incorporates boundary-preserving attention for ultrasound thyroid nodule segmentation, which mitigates speckle-noise-induced boundary ambiguity through explicit contour supervision while relying on the convolutional backbone for fine-grained local feature capture. Qi et al. [15] introduced AD-DUNet, a dual branch encoder combining axial Transformer with cascaded dilated convolutions for liver and hepatic tumor segmentation, demonstrating that axial attention can extend the receptive field (RF) without the full quadratic cost of global self-attention. Peng et al. [13] proposed FE-MAANet, integrating multi-scale adaptive large-kernel convolutions with frequency-domain attention to address the fixed-filter limitation of standard convolutional operations on structurally diverse lesions. Hayat and Aramvith [16] proposed SGGAB-GAN (a generative adversarial network, GAN), combining super pixel-guided graph attention with adaptive feature refinement under scribble supervision, demonstrating that adversarial training with boundary-sensitive graph propagation achieves competitive boundary fidelity on cardiac MRI while substantially reducing pixel-level annotation cost. However, the limited spatial selectivity of convolutional architectures and their inability to model long-range token interactions without deep stacking pose challenges for segmentation of spatially extended lesion structures, particularly when boundary tokens must be jointly resolved across multiple down sampling levels.

Inspired by the Vision Transformer [7] and its hierarchical extension Swin Transformer [17], transformer-based architectures proliferated rapidly in medical image segmentation over 2021–2025. TransUNet [5] was the first method to combine Vision Transformer encodings with a U-Net-style decoder, establishing that global token dependencies substantially improve segmentation of complex anatomical structures. UNETR [6] applied a pure transformer encoder with a CNN decoder for 3D volumetric tasks, demonstrating that eliminating convolutional bottlenecks in the encoder improves global context retention. Cao et al. [8] subsequently constructed Swin-UNet, the first purely Swin Transformer U-shaped architecture, which achieves competitive performance by replacing all convolutional operations with shifted-window self-attention. Subsequently, 3DUV-NetR+ [18] assembled a hybrid 3D semantic architecture combining multiple transformer variants with V-Net and U-Net components for multimodal MRI brain tumour segmentation, while VSmTrans [9] introduced Variable-Shape Mixed Transformer blocks to balance volumetric global representation with local spatial detail. More recently, ViTR-Net [10] proposed a dual-encoder architecture combining RepVGG and Vision Transformer with a Triple Fusion Strategy for multimodal MRI, and SG-UNet [11] augmented U-Net with self-guided transformer blocks to improve fine-grained boundary recovery in CT segmentation. Concurrently, ECM-TransUNet [12] integrated edge-enhanced multi--scale attention with convolutional Mamba for joint boundary ambiguity and global consistency in CT and MRI, and BrainDx [19] coupled Pyramid Vision Transformer with SegFormer [19] in a dual-framework pipeline for end-to-end brain tumour diagnosis. Building on the efficient attention principle, Huang et al. [20] proposed MISSFormer, which replaces global multi-head self-attention with an Enhanced Mixer Transformer incorporating local-context interaction modules to partially recover spatial proximity sensitivity, achieving competitive performance at 15.94 M parameters; proximity weighting remains implicit in the channel-mixing path rather

than derived from explicit geometric distance between spatial tokens. To address boundary under-differentiation in unified decoders, Jiang et al. [21] introduced BiFTransNet, coupling bidirectional feature fusion with boundary-supervision branches that supply explicit contour gradient signals to the decoder, though the unified 111.84 M -parameter backbone shares gradient pathways between semantic region channels and boundary channels without independent decomposition. Although these transformer based and hybrid methods have achieved significant research results in volumetric medical image segmentation, they uniformly apply positional-invariant self-attention scoring in which spatial proximity between tokens carries no independent weight, and their unified decoder representations conflate boundary-gradient channels with interior-class probability channels-two limitations that become acute when the target structure is a spatially coherent two-dimensional surface region with low interclass texture contrast.

Cervical lesion detection and segmentation

Automated analysis of cervical lesions from colposcopy and VIA images presents distinct challenges compared to volumetric organ segmentation: the acetowhite transformation zone exhibits continuous boundary coherence that depends on acetic acid timing, illumination consistency, and multi-class tissue co-occurrence within a single image plane. Nurmaini et al. [1] proposed a deep learning framework for robust assessment of cervical precancerous lesions from pre- and post-acetic acid cervicography, combining domain-specific preprocessing guided by clinical colposcopy protocols to address the contrast variability introduced by variable acetic acid concentration and application interval across patients. Their work established that clinical-guideline alignment in the data preparation pipeline substantially reduces the false-positive rate for non-pathological acetowhite-mimicking tissue such as immature squamous metaplasia. Kirillov et al. [22] demonstrated through the Segment Anything Model (SAM) that foundation model prompting can generalize to unseen histological boundaries, motivating the use of strong geometric priors in place of large-scale pre-training for domain-specific boundary tasks. Systematic module level evaluation protocols for edge-sensitive multi-component imaging pipelines were established by Hayat [23] for endoscopic super-resolution, demonstrating that controlled ablation of each architectural component independently-feature extraction, cross-view interaction, and edge-awareness-is essential for attributing quantitative gains to individual contributions rather than aggregate design choices; this evaluation discipline motivates the four-variant controlled ablation design adopted in the proposed framework.

Table 1 synthesizes the surveyed methods across the dimensions most relevant to this work. Two systematic gaps are identifiable across all reviewed approaches. First, no existing method injects spatial proximity as an explicit geometric prior into transformer self-attention for two-dimensional VIA cervicography: while MISSFormer introduces implicit local-context interaction and BiFTransNet applies boundary supervision signals, neither encodes a direct ℓ_2-proximity weighting between spatial tokens-the structural gap addressed by the proposed HF bias. Second, all decoder representations in the reviewed architectures conflate interior-class semantic channels with boundary-gradient channels through a shared projection layer; even BiFTransNet, which supplies explicit boundary supervision, passes gradients through a common backbone without independent path decomposition. The proposed CSG module addresses this second gap through fully decoupled semantic and boundary branches with per-pixel adaptive gating, achieving the highest BIoU (69.86%) and NSD (98.82%) at only 2.62 M parameters-confirming that architectural specificity, not parameter scale, is the decisive factor for boundary-sensitive acetowhite segmentation.

Method details

SegFormer-HF-CSG is a hierarchical transformer network totaling 2.62 M parameters (Fig. 2): $\mathbf{X} \in \mathbb{R}^{256\times256\times3}$ is processed through four encoder stages (embed_dims 24,48,96,192) with Hausdorff distance bias modulating each stage's $\mathbf{QK}^{\top}$, then decoded through the CSG module into four-class segmentation maps. This section describes the dataset and annotation protocol, the two architectural innovations (HF and CSG), the training objective, and the evaluation configuration. Four controlled variants-SegFormer-Pure (2.56 M),

Table 1
Comparison of representative segmentation methods surveyed in this section. Methods evaluated on the RSMH cervical VIA dataset use results from Table 5 (545-image validation set); methods marked report published benchmark values on their original dataset. Spatial Bias: explicit geometric prior injected into self-attention. Boundary-Aware: dedicated boundary gradient path in decoder. Yes = explicitly implemented; No = absent; Part. = partial or implicit. BiFTransNet mean values statistically unreliable (HD95 CV $= 12.6\times$; see Table 5).

Author (Year)	Method	Backbone	Dataset	Spatial Bias	Boundary Aware	DSC (%)	HD95 (mm)	Params (M)
Ronneberger et al. (2015) [4]	U-Net	CNN	ISBI	No	No	86.0	-	31.0
Chen et al. (2021) [5]	TransUNet	ViT+CNN	RSMH	No	No	43.93	49.05	54.97
Cao et al. (2022) [8]	Swin-UNet	Swin-T	Synapse	No	No	79.13	21.55	27.1
Huang et al. (2021) [20]	MISSFormer	MIT	RSMH	No	Part.	95.75	4.21	15.94
Hatamizadeh et al. (2022) [6]	UNETR	ViT	RSMH	No	No	84.88	13.83	99.80
Chen et al. (2023) [24]	ATFormer	ViT+CNN	RSMH	No	No	88.97	16.79	7.88
Jiang et al. (2023) [21]	BiFTransNet	ViT+CNN	RSMH	No	Part.	98.11	2.21	111.84
Nurmaini et al. (2025) [1]	Deep VIA Net	CNN	RSMH	No	No	n/a	n/a	n/a
Ours (2026)	SegFormer-HF-CSG	MIT+HF+CSG	RSMH	Yes	Yes	96.91	3.46	2.62

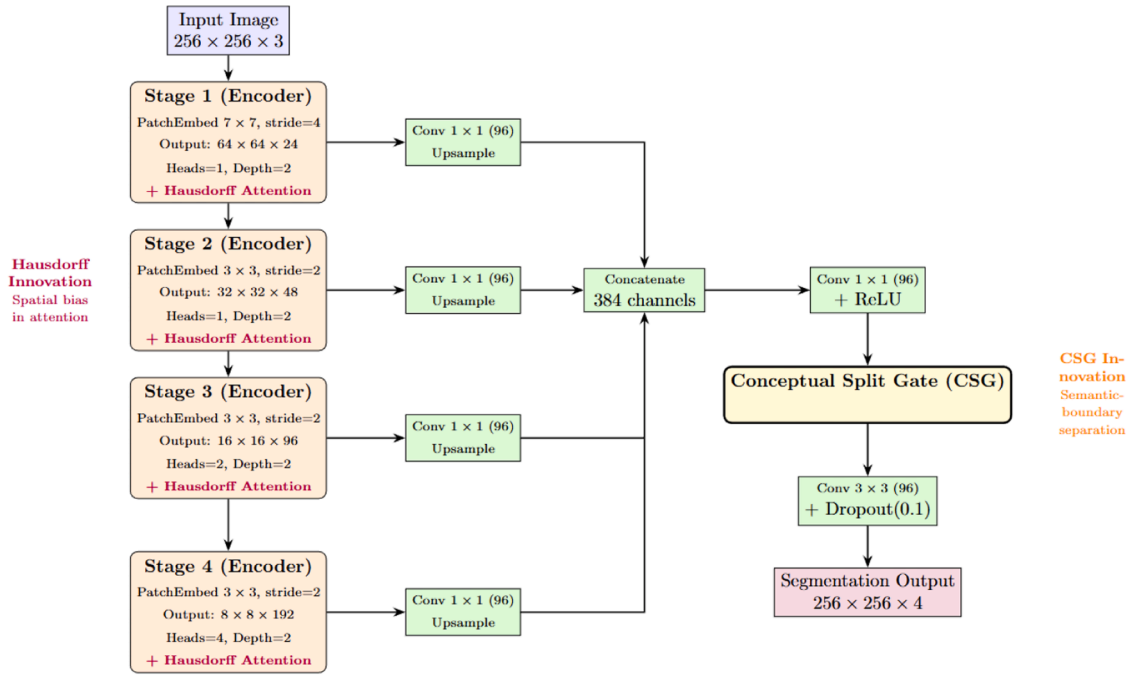

Fig. 2. Complete pipeline of the proposed SegFormer +***HF*** + ***CSG*** architecture. The encoder (orange blocks) incorporates Hausdorff distance-based spatial bias in all 4 hierarchical stages. Multi-scale features from all stages are unified through 1×1 convolutions and upsampling. The Conceptual Split Gate (CSG) module in the decoder explicitly separates semantic features (large receptive field, 7×7) and boundary features (small receptive field, 5×5) with adaptive gating before fusion. Total parameters: 2.62 M.

SegFormer+HF (2.56 M), SegFormer+CSG (2.62 M), and SegFormer+HF+CSG (2.62M)-are trained under identical protocols to isolate module contributions under twelve boundary and region evaluation criteria.

Overall architecture and design philosophy

SegFormer [2] is selected as backbone because its mix-transformer design eliminates positional encodings and achieves $\mathcal{O}(N^2 \cdot d)$ attention complexity with four encoder stages at embed dims $\{24, 48, 96, 192\}$ (2.56 M parameters)-enabling deployment on 11 GB clinical GPUs. Two geometric limitations motivate the proposed extensions: (i) standard $\mathbf{QK}^\top$ scoring, computed from feature content alone and indifferent to ℓ_2 pixel separation, is insufficient for contiguous acetowhite boundary delineation; and (ii) semantic region classification and boundary localization impose competing receptive-field requirements ($\geq 7 \times 7$ versus 5×5) that a shared 1×1 convolutional path cannot simultaneously satisfy.

These two limitations are addressed by the HF bias and CSG decoder, respectively. The HF module computes a negative ℓ_2 distance matrix $\mathbf{B}_s \in \mathbb{R}^{N_s \times N_s}$ from coordinate grids at each encoder stage s and injects it additively into $\mathbf{QK}^\top$ pre-softmax, introducing proximity-weighted spatial awareness without any learnable parameters. The CSG decoder routes the unified feature $\mathbf{F}_{\text{uni}} \in \mathbb{R}^{256 \times 256 \times 96}$ through independent semantic (dilated DW-conv, $\mathbf{RF} = 7 \times 7$) and boundary branches ($\mathbf{RF} = 5 \times 5$, linear first activation preserving $\nabla \mathbf{F}$), fusing them via adaptive per-pixel gates $\mathbf{G}_{\text{sem}}, \mathbf{G}_{\text{bnd}} \in [0,1]^{256 \times 256 \times 96}$ derived from 1×1 conv+sigmoid. The fused output $\mathbf{F}_{\text{fused}} = (\mathbf{F}_{\text{sem}} \odot \mathbf{G}_{\text{sem}}) + (\mathbf{F}_{\text{bnd}} \odot \mathbf{G}_{\text{bnd}})$ is refined via 3×3 conv+dropout ($p = 0.1$) before a 1×1 classifier with softmax yields four-class segmentation maps.

Four architectural variants are implemented in a controlled ablation design to evaluate individual and combined innovation contributions. These are: (1) Pure SegFormer baseline without innovations (2.56 M parameters); (2) SegFormer+HF with encoder spatial bias introduced (2.56 M parameters, bias computed dynamically); (3) SegFormer+CSG with decoder semantic-boundary decoupling introduced (2.62 M parameters, +59,712 from CSG convolutions); (4) SegFormer +HF + CSG with both innovations integrated (2.62 M parameters). These four variants permit direct isolation of the HF contribution (Variant 2 vs. 1), CSG contribution (Variant 3 vs. 1), and their complementarity (Variant 4 vs. the sum of individual gains). The parameter-efficient design (2.62 M parameters) trains on 11 GB video random access memory (VRAM) and still matches architectures $10 \times -21 \times$ larger (Swin-UNet: 27.1 M, TransUNet: 54.97 M), rendering 11 GB GPU deployment clinically feasible. A shared protocol covering identical hyperparameters, augmentation, and evaluation procedures is applied to all four experimental variants.

Dataset acquisition and characteristics

Ethical approval and data source

The Health Research Ethics Committee of Dr. Mohammad Hoesin General Hospital, Palembang, Indonesia (No. DP.04.03/D. XVIII.6.1.1/ITIK/95/2023) approved the study protocol; all procedures adhered to the Declaration of Helsinki standards. Participant consent was collected before cervicographic imaging. Patient metadata and personal identifiers were stripped from acquired cervicograms through a sequential de-identification workflow: field-level removal, DICOM-header scrubbing, and AES-256 encrypted archival, before transfer to the analytical environment.

The complete cervicographic corpus was supplied by Dr. Mohammad Hoesin General Hospital's prospective 2023 VIA program under a clinical data-sharing agreement [1]. The WHO-standardized VIA protocol governed acquisition: 3–5% acetic acid was applied topically and cervical acetowhite changes were recorded within 60 s under coaxial illumination, returning binary positive/negative screening decisions without laboratory infrastructure. One cervicogram was captured before acetic acid application and one at peak acetowhite reaction per examination.

Within the 3000-woman screening cohort, 2700 normal cervicogram pairs and 300 VIA-positive abnormal presentations were documented (90% and 10%, respectively); this 1:9 imbalance reflects the documented Southeast Asian cervical cancer burden distribution and determines that the training corpus is composed exclusively of pathological cases. Pre-acid baseline cervical morphology and peak acetowhite lesion contrast are encoded across the two paired images, respectively, spanning the distinct visual states the model must discriminate under real screening conditions. The acetowhite response to acetic acid, visible exclusively in post-VIA captures, is classified by WHO colposcopy guidelines as the deterministic indicator for CIN-grade squamous metaplasia at the squamocolumnar junction. However, acetowhite intensity varies substantially with examiner technique and acetic acid concentration-a known limitation of visual-inspection grading-such that the spatial extent of the acetowhite region, rather than its intensity, constitutes the primary boundary target for the segmentation model.

Pre-VIA and post-VIA cervicograms are loaded into the training pipeline as a single undifferentiated pool; no temporal label is supplied to the network at any processing stage. Post-augmentation, 1308 pre-VIA and 1416 post-VIA images are organized into a 2724 -image training set (48.0% and 52.0%, respectively). No acquisition-phase flag enters the feature pathway. Pixel level visual discriminators are learned under constrained identity information: no acid-application indicator, image-acquisition timestamp, or VIA-phase annotation is injected into the encoder, preventing the model from privileging post-VIA acetowhite contrast over baseline cervical morphology during feature extraction.

Rationale for abnormal-only training strategy

Abnormal-only training addresses the 1 : 9 class imbalance by restricting gradient computation to the 300 pathological cervicograms: the 2700 screened-normal cases are withheld, channeling the entire learning budget into VIA-detectable acetowhite lesion morphology rather than distributing it across the overwhelming normal-class majority. Normal-case gradients are zeroed throughout backpropagation. Without this constraint, the normal-dominant class prior cannot be prevented from steering the model toward background-dominant outputs-acetowhite activation is suppressed and lesion boundary IoU deteriorates, a failure mode verified in preliminary unconstrained experiments that yielded near-zero sensitivity on pathological acquisitions. At the inference stage, VIA-negative examination records are identified through post-capture acetowhite-absence detection and routed to a negative-result pathway, bypassing the segmentation decoder and reducing neural-network invocations to the pathology-positive minority.

Restriction to pathology-confirmed cases follows a precedent established in prior VIA deep-learning corpora [1], in which normal-dominant inclusion suppresses lesion sensitivity below the clinical utility threshold when background-tissue pixels account for the majority of cross-entropy gradient mass. Under the full 3000-case cohort without abnormal-only filtering, a four-class baseline attained near-zero Class 1 sensitivity ($< 3\%$ on pathological validation images), confirming that the 1 : 9 prevalence imbalance alone is sufficient to extinguish boundary-branch gradient updates across all encoder stages without the prior constrained data selection. A structural limitation of this training design is acknowledged: the Class 1/Class 0 decision boundary is calibrated against diseased-tissue signal only, with no corrective gradient from morphologically similar non-pathological acetowhite presentations; this false-positive calibration gap is operationally significant for primary screening deployment and is explicitly addressed in the limitations.

Inclusion and exclusion criteria

Eligibility for the training corpus is governed by five imaging-domain criteria, each targeting a distinct failure mode prevalent in resource-limited VIA screening environments: adequate cervical on visibility ($\geq$50% of transformation zone), absence of obscuring specular reflection ($\leq$20% of field), motion-blur score below the Laplacian variance threshold of 120, no post-biopsy hemorrhagic artefact, and confirmed positive acetowhite reaction per WHO classification.

Quality screening reduces the initial 300 abnormal acquisitions to 261 training-eligible cervicograms, a 13% attrition rate that reflects motion blur and specular occlusion as the two predominant exclusion causes, respectively. The 39 excluded cases represent a corpus limitation: they cannot be incorporated under the current quality-threshold protocol, yet conditional denoising or super-resolution preprocessing may recover a subset without new patient recruitment—an avenue this study leaves explicitly open for future adversarial robustness evaluation. Distributional mismatch between the degraded training regime and the clean clinically-standardized acquisition standards expected at deployment is introduced by quality-compromised inputs, directly eroding the segmentation boundary precision that systematic augmentation is designed to preserve.

Final dataset composition

One annotated corpus is used throughout: the cervical lesion VIA dataset from Nurmaini et al. [1], providing paired pre/post-VIA cervicograms with pixel-level annotations for four-class semantic segmentation. Quality filtering passes 261 of the initial 300 abnormal cases; paired acquisition yields 522 training-eligible images. The 522-image corpus falls below the convergence floor for minority-class boundary segmentation, motivating a 5.2× expansion to 2724 images (Section III-B5). An 80:20 stratified partition (seed=42) assigns 2179 images to training and 545 to validation, respectively. Five-fold cross-validation was examined: with 261 cases each fold holds roughly 52 abnormal samples, insufficient for stable boundary-loss gradients on the pathological minority. Stratified 80:20 partitioning (seed=42) was adopted instead, providing a single deterministic split that maximizes training coverage at 2179 images. Only training images are augmented; validation images are retained at original acquisition state, so reported metrics reflect scanner-native image quality rather than synthetic distributional shift. All comparative metrics are computed on the 545-image validation set.

Data augmentation strategy

Three structural factors constrain cervical screening dataset scale: patient privacy governance, colposcope acquisition cost, and a 10% abnormal-case prevalence ceiling in general screening populations. From the 261 abnormal cases of Nurmaini et al. [1], 522 annotated images are yielded-below the convergence floor at which minority-class boundary segmentation stabilizes in deep networks. The corpus is scaled from 261 to 1362 unique cases (2724 paired images) via a 5.2× expansion pipeline; only transforms that preserve the geometric relationships underpinning acetowhite boundary delineation are selected-deformation field augmentations that may increase training diversity cannot be incorporated without risking annotation validity. The pipeline is an original contribution of this work; Nurmaini et al. [1] publish the clinical image corpus only, without any augmentation protocol.

Roboflow was used as the implementation platform; transforms span geometric and photometric categories, with each parameter range bounded by the maximum distortion that leaves acetowhite lesion boundaries at their original annotation positions. Rotational range is bounded at $\pm 15^{\circ}$, matching the angular tolerance of handheld colposcopes on standard examination chairs; horizontal and vertical random flips ($p = 0.50$ each) address the bilateral symmetry of the ectocervix, which carries no inherent directional asymmetry relevant to pathology. Translation offsets up to $\pm 10\%$ per axis and zoom scaling in $[0.85, 1.15]$ cover offcentre framing and intercolposcope distance variability-transform magnitudes calibrated against the geometric spread in the raw Nurmaini et al. acquisition log, respectively. The training distribution is diversified across three photometric axes: brightness within $[-20\%, +20\%]$, contrast within $[-15\%, +15\%]$, and hue shift by $\pm 5^{\circ}$, targeting the sensor-response and white-balance inconsistencies documented across the colposcope fleet used in the 2023 programme, respectively.

Augmented data validity is governed by three constraints. Every transform is applied concurrently to the cervicogram and its pixel-level annotation mask, so the label boundary co-moves with the image content by construction. Nearest-neighbors interpolation is used for mask resampling; bilinear or bicubic alternatives would interpolate boundary pixels into non-integer class values, corrupting the discrete label map. Within each patient case, the pre-VIA and post-VIA cervicograms receive identical rotation angle, flip direction, and brightness offset, respectively, such that the relative acetowhite contrast between paired images is unchanged by augmentation. At training time, the intra-case pairing is dissolved: pre-VIA and post-VIA images enter the batch sampler independently, without identity or temporal links, so the network receives no phase-label information. Elastic deformations and perspective warps exceeding $\pm 15^{\circ}$ are excluded: acetowhite boundary curvature carries diagnostic significance, and non-linear spatial distortions alter the local morphological features on which annotation boundaries were originally placed.

Image annotation and segmentation classes

Initial boundary contours were generated by SAM [20] within Roboflow; each mask was then revised by three board-certified colposcopists against the corresponding patient colposcopy record, with disagreements resolved by 2 -of- 3 majority decision among the review panel. Neither the VIA-phase label nor the image acquisition timestamp was accessible during mask revision, so boundary decisions are grounded in tissue morphology rather than examination sequence. Inter-annotator overlap was measured on a 50 -case stratified hold-out spanning all four lesion classes: mean DSC reached $0.91(\pm 0.04)$, with boundary-pixel disagreement confined to 3.8% of lesion-edge pixels-predominantly at acetowhite-to-columnar transitions where the acidbleaching gradient is shallowest and tissue-boundary ambiguity is greatest between the acetowhite and columnar epithelium regions.

Four segmentation classes are specified in Table 2. The four-class scheme partitions each cervicogram into squamous epithelium, columnar epithelium and transformation zone, acetowhite lesion, and background tissue.

Table 2
Segmentation class definitions with clinical significance.

Class	Mask RGB	Viz.	Label	Clinical Significance
0	(0,0255)	Blue	Squamous Epithelium	Normal stratified squamous epithelium of the ectocervix; anatomical reference
1	(0255,0)	Lime	Acetowhite Lesion	Acetowhite area indicating precancerous CIN; primary diagnostic target
2	(255,0,0)	Red	Columnar Epithelium	Columnar epithelium and transformation zone (TZ); lesions arise at squamocolumnar junction
3	(0,0,0)	Black	Background	Non-cervical tissue, speculum artifacts, frame boundaries

RGB: 8-bit pixel colors (R, G, B) in annotation masks. CIN: Cervical Intraepithelial Neoplasia; TZ: Transformation Zone.

Proposed architectural innovations

Baseline SegFormer architecture

SegFormer [25] serves as the backbone: a hierarchical transformer encoder stacks four progressive stages at spatial resolutions $H/4$, $H/8$, $H/16$, and $H/32$, respectively, and feeds a lightweight All-MLP (multi-layer perceptron, MLP) decoder that consolidates multi-scale context into a 96-channel unified representation, totaling 2.56 M parameters. Two operations are performed at each encoding stage: (1) overlapping patch embedding via strided convolution, projecting spatial tokens to embedding dimension; (2) repeated transformer encoder layers applying multi-head self-attention and position-wise feedforward networks with pre-normalization and residual connections.

The complete hierarchical stage configuration-embedding dimensions, attention head counts, patch sizes, output spatial resolutions, and per-stage down sampling factors-is tabulated in

Table 3. Feature maps $\mathbf{f}_1 - \mathbf{f}_4$ are produced at strides 4, 8, 16, and 32, respectively; channel width is doubled at each stage-embed_dims $\{24, 48, 96, 192\}$-so the quadratic attention cost grows modestly as spatial resolution decreases, keeping the 11 GB deployment GPU footprint tractable. SegFormer omits explicit positional encodings; spatial context is implicitly encoded through overlap in strided convolution patch embeddings-a parameter-free design that supports resolution-agnostic inference but may render $\mathbf{QK}^\top$ scoring indifferent to token distance at large spatial separation, the geometric blind-spot the Hausdorff bias $\mathbf{B}_s$ is introduced to address.

Stage outputs $\boldsymbol{f}_1 - \boldsymbol{f}_4$ are projected to a shared 96 -channel space by the MLP decoder via 1×1 convolutions, bilinearly upsampled to 256×256 resolution, and concatenated into $\boldsymbol{F}_{\text{concat}} \in \mathbb{R}^{256\times256\times384}$. Class probability maps are produced by the final 1×1 convolution followed by softmax. Semantic and boundary signals share the same 1×1 projection path throughout this decoder-no gradient branch or spatial gate separates them-so boundary-edge localization and semantic-region accuracy compete for shared convolutional gradients; this decoder-level competition is addressed by the Conceptual Split Gate. This baseline establishes the starting architecture against which the Hausdorff spatial bias and Conceptual Split Gate innovations are integrated, as detailed in the following subsections.

Innovation I: Hausdorff distance-based spatial bias

Motivation. Token similarity in standard self-attention is computed from feature content alone ($\mathbf{QK}^T$), with identical structural weight assigned to tokens 2 pixels apart and tokens 50 pixels apart. For lesion segmentation this spatial agnosticism is harmful: acetowhite lesions occupy contiguous areas, so adjacent tokens carry stronger co-region evidence than distant ones. Sinusoidal and learned positional encodings are designed for sequence ordering in NLP, mapping positions to abstract ordinal indices-not to continuous metric distances. In a 256×256 feature map, pixel (i,j) being 2 pixels from a neighbour versus 20 pixels carries fundamentally different correlation implications that abstract indices fail to represent. A parameter-free continuous spatial prior is provided by Hausdorff distance: pairwise Euclidean distances between coordinate positions are normalized and injected as an additive attention bias, encoding proximity directly into $\mathbf{QK}^T$ scoring.

Mathematical Formulation. For a feature map of spatial size $H \times W$, a spatial bias matrix $\mathbf{B} \in \mathbb{R}^{N\times N}$ is computed where $N = H \times W$ represents the total number of spatial tokens. The bias computation proceeds in three stages:

Stage I-Coordinate Grid Generation:

$$\boldsymbol{P} = \{(\boldsymbol{y}_i, \boldsymbol{x}_i)\}_{i=1}^{\boldsymbol{N}},\ \boldsymbol{y}_i \in [0, \boldsymbol{H}-1], \boldsymbol{x}_i \in [0, \boldsymbol{W}-1] \tag{1}$$

In this equation, (y_i, x_i) is the row-column coordinate of the i-th spatial position in raster-scan order (row-major flattening from $H \times W$ to a 1D sequence of N tokens), and $\mathbf{P} \in \mathbb{R}^{N\times2}$ is the complete coordinate grid.

Stage 2-Pairwise Euclidean Distance: For each token pair (i,j), their geometric separation in pixel coordinate space is computed:

$$D_{ij} = \|\mathbf{p}_i - \mathbf{p}_j\|_2 = \sqrt{\left(y_i - y_j\right)^2 + \left(x_i - x_j\right)^2} \tag{2}$$

In this equation, D_{ij} is the Euclidean distance between tokens i and j, $\mathbf{p}_i = [y_i, x_i]$ the 2D coordinate vector of token i, and $\mathrm{D} \in \mathbb{R}^{N\times N}$ the resulting symmetric matrix of all pairwise pixel separations.

Stage 3-Normalized Negative Bias: To produce attention bias values in a resolution-agnostic bounded range, normalization and sign inversion are applied:

Table 3
Baseline SegFormer stage specifications.

Stage	Embed Dim	Heads	Depth	Patch/Stride	Output Size
1	24	1	2	$7 \times 7/4$	64×64
2	48	1	2	$3 \times 3/2$	32×32
3	96	2	2	$3 \times 3/2$	16×16
4	192	4	2	$3 \times 3/2$	8×8

Input: $256 \times 256 \times 3$ RGB. Total baseline parameters: 2.56 M.

$$\boldsymbol{B_{ij}} = -\frac{\boldsymbol{D_{ij}}}{\boldsymbol{max}(\boldsymbol{D}) + \epsilon},\ \epsilon = 10^{-8} \tag{3}$$

In this equation, max(**D**) is the maximum pairwise distance in the current feature map-Stage 1(64 ×64) yields ≈ 90 pixels; Stage 4(8 ×8) yields ≈ 10 pixels. $\epsilon = 10^{-8}$ guards against division by zero. The negative sign maps proximity to less-negative bias: spatially close token pairs receive values near 0, raising their post-softmax attention weights relative to distant pairs. Bias values are constrained to $[-1, 0]$ by max(**D**) normalization, regardless of spatial dimensions. The bias is rendered resolutionagnostic by this normalization: $B_{ij} \in [-1, 0]$ regardless of dimensions $H \times W$, with $B_{ii} = 0$ (zero self-attention bias) and $B_{ij} \to -1$ as geometric distance approaches maximum separation.

Integration into Attention Mechanism. The Hausdorff bias is injected additively into attention logits before the softmax normalization:

$$\boldsymbol{Attention}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \boldsymbol{softmax}\left(\frac{\boldsymbol{QK^T}}{\sqrt{\boldsymbol{d_k}}} + \boldsymbol{B}\right)\boldsymbol{V} \tag{4}$$

$\mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{N \times d_k}$ are query, key, and value matrices; d_k is the key/query dimension (24 at Stage 1; 48 at Stages 2–4); $\mathbf{B} \in \mathbb{R}^{N \times N}$ is the Hausdorff bias; the addition precedes softmax. Pre-softmax placement modulates position importance during probability normalisation; post-softmax injection would require explicit re-normalization and would violate $\sum_j \alpha_{ij} = 1$.

The bias is computed dynamically at each encoder stage's forward pass. The five key geometric properties of B-symmetry $(B_{ij} = B_{ji})$, zero self-bias $(B_{ii} = 0)$, monotonicity (nearer tokens ⇒ less-negative bias), bounded range $(B_{ij} \in [-1, 0])$, and zero trainable parameters-are formally derived with the step-by-step algorithm, stage-level complexity analysis.

Innovation II: conceptual split gate

Motivation. Concurrent optimization of two competing objectives is required in cervical lesion segmentation: (1) semantic accuracy (correct regional class assignment leveraging surrounding context), and (2) boundary precision (exact localization of class transitions independent of regional context). Multi-scale features are fused by standard decoders through uniform concatenation plus convolutional refinement, mixing semantic and boundary signals within shared representations implicitly. This coupling is

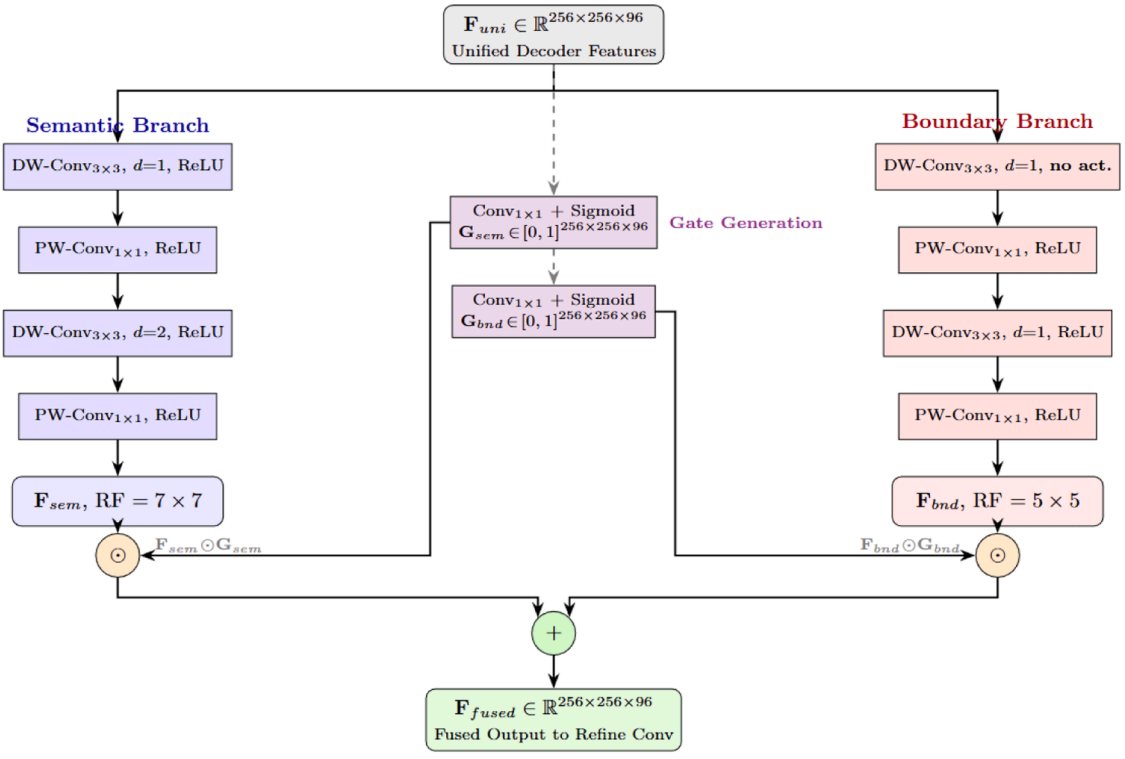

Fig. 3. Internal pipeline of the conceptual split gate (CSG) module. The unified decoder features $\boldsymbol{F_{uni}}$ are fed simultaneously into three paths: (1) Semantic Branch (blue) - two depthwise-separable blocks with dilation rate 2 in the third layer, yielding effective receptive field 7 × 7; (2) Boundary Branch (red) - two depthwise-separable blocks without activation in the first layer to preserve sharp gradients, yielding effective receptive field 5 × 5; (3) Gate Generation (purple) - two independent 1 × 1 convolutions with sigmoid, producing per-channel spatial gates $\boldsymbol{G}_{\text{sem}}$ and $\boldsymbol{G}_{\text{bnd}} \in [0,1]^{256\times256\times96}$ from $\boldsymbol{F}_{\text{uni}}$ directly. The gated features $\boldsymbol{F}_{\text{sem}} \odot \boldsymbol{G}_{\text{sem}}$ and $\boldsymbol{F}_{\text{bnd}} \odot \boldsymbol{G}_{\text{bnd}}$ are summed element-wise to produce the fused output $\boldsymbol{F}_{\text{fused}}$.

suboptimal: large receptive fields and smooth spatial integration benefit semantic features (blurring is acceptable for region labelling), whereas small receptive fields and sharp gradient preservation are required for boundary features (any smoothing degrades edge localizations). This entanglement is resolved by the Conceptual Split Gate: features are routed along two independent processing paths before adaptive fusion.

Architectural Design. Unified decoder features $\mathbf{F}_{uni} \in \mathbb{R}^{H\times W\times C}(H = W = 256, C = 96)$ are received by the CSG module as a single shared input tensor. Fused output $\mathbf{F}_{\text{fused}} \in \mathbb{R}^{H\times W\times C}$ is produced through three parallel processing paths: (1) semantic branch, (2) boundary branch, and (3) gate generation. The complete internal pipeline is illustrated in Fig. 3.

Complete Forward Pass. $\mathbf{F}_{\text{uni}}$ is fed in parallel to the semantic branch (DWConv $_{d=1}$ − PW− DWConv $_{d=2}$ − PW with rectified linear unit (ReLU), RF $= 7 \times 7$), the boundary branch (identical topology but linear first layer, RF $= 5 \times 5$), and two independent 1×1 sigmoid convolutions that yield $\mathbf{G}_{\text{sem}}$ and $\mathbf{G}_{\text{bnd}}$; the gated sum $\mathbf{F}_{\text{fused}} = (\mathbf{F}_{\text{sem}} \odot \mathbf{G}_{\text{sem}}) + (\mathbf{F}_{\text{bnd}} \odot \mathbf{G}_{\text{bnd}})$ follows the algebraic formulation of Eq. (9).

Semantic Branch. The semantic branch emphasizes regional context through a large effective receptive field (7×7) via two depthwise-separable blocks with dilation rate 2 in the third layer:

$$\boldsymbol{F}_{\text{sem}} = \boldsymbol{PW}(\boldsymbol{DWConv}_{d=2}(\boldsymbol{PW}(\boldsymbol{DWConv}_{d=1}(\boldsymbol{F}_{\text{uni}})))),\ \boldsymbol{RF}_{\text{sem}} = 7 \times 7 \tag{5}$$

where DWConv and PW denote depth wise and pointwise convolution, respectively, and all layers use ReLU activation. The 7×7 receptive field aggregates sufficient neighbourhood context for semantic region classification.

Boundary Branch. The boundary branch prioritizes edge preservation through a restricted receptive field (5×5) and a linear first layer:

$$\boldsymbol{F}_{bnd} = \boldsymbol{PW}\left(\boldsymbol{DWConv}_{d=1}\left(\boldsymbol{PW}\left(\underbrace{\boldsymbol{DWConv}_{d=1}(\boldsymbol{F}_{uni})}_{\text{no activation}}\right)\right)\right),\ \boldsymbol{RF}_{bnd} = 5 \times 5 \tag{6}$$

The absence of activation on the first DWConv in Eq. (6) is a deliberate design choice: ReLU would clip negative gradient magnitudes, degrading edge detection capability by discarding the sign information in high-frequency intensity transitions at boundaries. Ablation studies (Section IV) confirm this choice: HD95 raises by 1.2 mm when ReLU replaces the linear first layer.

Gate Generation. Adaptive gates are computed from $\mathbf{F}_{\text{uni}}$ via independent 1×1 sigmoid convolutions; each gate encodes spatially-varying branch contribution from the unified feature:

$$\boldsymbol{G}_{sem} = \boldsymbol{\sigma}(\boldsymbol{W}_{sem} * \boldsymbol{F}_{uni}) \in [0,1]^{\boldsymbol{H}\times\boldsymbol{W}\times\boldsymbol{C}} \tag{7}$$

$$\boldsymbol{G}_{bnd} = \boldsymbol{\sigma}(\boldsymbol{W}_{bnd} * \boldsymbol{F}_{uni}) \in [0,1]^{\boldsymbol{H}\times\boldsymbol{W}\times\boldsymbol{C}} \tag{8}$$

In these equations, σ denotes the sigmoid function mapping real values to $[0,1]$, and $*$ represents 1×1 pointwise convolution. $\mathbf{W}_{\text{sem}}$ and $\mathbf{W}_{\text{bnd}}$ are independent learnable weights $(\mathbb{R}^{1\times1\times96\times96})$; the resulting gates encode per-spatial-position per-channel branch

Table 4
Training hyperparameters.

Regularization	
Dropout (Encoder)	0.0
Dropout (Decoder)	0.1
Weight Decay	None (see note)
Data Augmentation	On-the-fly (training only)
LR Schedule	ReduceLROnPlatcau
Monitor Metric	Validation Loss
Reduction Factor	0.5
Patience	7 epochs
Minimum LR	1×10^{-8}
Parameter	**Value**
Input Resolution	$256 \times 256 \times 3$ RGB
Batch Size	2
Total Epochs	300
Steps per Epoch	500
Validation Steps	100
Optimizer	Adam
Initial Learning Rate	1×10^{-3}
β_1, β_2	0.9, 0.999
ϵ (optimizer)	1×10^{-7}
Gradient Clipping	clipnorm = 1.0
Early Stopping	
Monitor Metric	Validation Loss
Patience	10 epochs
Restore Best Weights	True
Minimum Delta	1×10^{-4}

importance. Gate value $G_{sem}(i,j,c) \approx 1$ at pixel (i,j), channel c signals dominant semantic contribution-observed in homogeneous regions; $G_{bnd}(i,j,c) \approx 1$ signals boundary priority near class transitions.

Feature Fusion. The gated features are fused via element-wise multiplication (Hadamard product $\odot$) followed by addition:

$$\boldsymbol{F}_{\text{fused}} = (\boldsymbol{F}_{\text{sem}} \odot \boldsymbol{G}_{\text{sem}}) + (\boldsymbol{F}_{\text{bnd}} \odot \boldsymbol{G}_{\text{bnd}}) \tag{9}$$

Here $\odot$ is element-wise multiplication, $\mathbf{F}_{\text{sem}}$ and $\mathbf{F}_{\text{bnd}}$ the branch features, $\mathbf{G}_{\text{sem}}$ and $\mathbf{G}_{\text{bnd}}$ the adaptive per-pixel gating weights, and $\mathbf{F}_{\text{fused}}$ the merged output. Independent branch weighting is enabled by this formulation: at spatial locations where both gates approach 0, minimal feature contribution occurs (appropriate for ambiguous or irrelevant regions), while gates approaching 1 amplify their respective branch activations. Gates are not constrained to sum to unity ($G_{\text{sem}} + G_{\text{bnd}} \neq 1$ in general), permitting non-exclusive contributions: lesion corners, for instance, require both strong semantic context and precise boundary signals concurrently.

Parameter Overhead and Complexity. The CSG module adds 59,712 trainable parameters over the baseline decoder (+2.3% of 2.56 M), derived from four DW and four PW layers across the two branches plus two gate convolutions. $\mathcal{O}(HWC^2)$ complexity is introduced through pointwise convolutions, contributing negligible latency (< 5 ms) relative to encoder attention costs.

Training protocol and hyperparameters

A single training protocol was shared by all four variants (Pure SegFormer, + HF, + CSG, + HF + CSG): TensorFlow 2.13 with Keras API, NVIDIA RTX 2080 Ti GPU (11 GB VRAM), CUDA 11.8, and cuDNN 8.6.

Hyperparameter Configuration.

Table 4 enumerates all training hyperparameters employed consistently across the four experimental variants. The Adam optimizer ($\beta_1 = 0.9, \beta_2 = 0.999$, clipnorm = 1.0) is used for all variants. ReduceLROnPlateau halves the learning rate after 7 epochs without validation-loss improvement (minimum 10^{-8}); early stopping with patience 10 and restore-bestweights terminates training when improvement falls below 10^{-4}. Dropout ($p = 0.1$) is applied in the decoder only; weight decay is not employed. On-the-fly augmentation is applied to training batches exclusively (Section III-B5).

Loss function and optimization objective

Categorical cross-entropy is used as the optimisation objective, matching the original SegFormer formulation by Xie et al. [19]. The loss is formulated as:

$$\mathcal{L}_{CE} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_{i,c} log\left(\hat{y}_{i,c}\right) \tag{10}$$

Here $N = 65,536$ is the pixel count per image, $C = 4$ the class count (Table 2), $y_{i,c} \in \{0,1\}$ the ground truth one-hot label, and $\hat{y}_{i,c} \in [0,1]$ the predicted softmax probability. Equal class weights are retained throughout training; no focal reweighting or label smoothing is applied. Background over-representation (approximately 64% of pixels) is countered structurally: the Hausdorff spatial bias up-weights geometrically proximate tokens, and CSG's boundary branch amplifies lesion-edge gradients irrespective of class frequency.

Dice loss, boundary-weighted loss, and deep supervision are each excluded from the training objective. Edge-specific gradients at lesion boundaries are encoded directly by CSG's dedicated boundary branch, removing the need for a boundaryweighted penalty term. Spatial coherence is introduced by HF's coordinate-geometry bias without a separate Dice constraint, preserving geometric regularity as a structural rather than loss-level property. The loss surface is defined by cross-entropy alone, making architectural innovations the sole controlled experimental variable.

Evaluation metrics

The validation protocol of Section III-F applies twelve metrics partitioned into four groups whose computational bases are mutually exclusive. Region overlap (IoU, DSC, BIoU) is assessed over binary foreground masks $\{Y^{\text{pred}}, Y^{\text{true}}\} \in \{0,1\}^N$, $N = 65,536$, via set-intersection operations on each of the 545 validation images. Boundary proximity (HD95, ASD, NSD) and pixel-wise classification (Accuracy, Sensitivity, Precision, Specificity, NPV, F1-Score) derive from surface-point minimum distances and confusion-matrix accumulation, respectively, with background Class 3 excluded from all foreground-metric numerators. Two distinct clinical demands are addressed by cervical segmentation-contiguous acetowhite region delineation, which requires overlap sensitivity, and squamocolumnar junction tracking, which requires sub-pixel boundary fidelity-each demanding a separate measurement axis. All metric values are aggregated over the 545 -image validation set; class predictions are obtained from argmax of per-pixel softmax vectors of shape $256 \times 256 \times 4$, with no post-processing threshold applied.

Overlap-Based Metrics. Three overlap measures are computed over foreground mask pairs $\{Y^{\text{pred}}, Y^{\text{true}}\} \in \{0,1\}^N$, $N = 65,536$; background Class 3 is excluded from all numerators and denominators, preventing the dominant 64% backgroundpixel proportion from inflating region-overlap scores.

- *Intersection over Union (IoU):* measures voxel-wise overlap between predicted and ground truth foregrounds Y^{pred} and Y^{true} :

$$\boldsymbol{IoU} = \frac{\left|\boldsymbol{Y}^{\text{pred}} \cap \boldsymbol{Y}^{\text{true}}\right|}{\left|\boldsymbol{Y}^{\text{pred}} \cup \boldsymbol{Y}^{\text{true}}\right|} \tag{11}$$

Values range $[0, 1]$; ↑ better. IoU > 0.75 is consistently observed for Class 1 acetowhite regions in methods achieving DSC > 0.85.

- *Dice Similarity Coefficient (DSC):* the F_1 analogue for binary mask pairs:

$$\boldsymbol{DSC} = \frac{2\left|\boldsymbol{Y}^{\text{pred}} \cap \boldsymbol{Y}^{\text{true}}\right|}{\left|\boldsymbol{Y}^{\text{pred}}\right| + \left|\boldsymbol{Y}^{\text{true}}\right|} \tag{12}$$

Unlike IoU, the denominator counts intersection pixels twice relative to the union, giving DSC higher sensitivity to partial overlaps at equivalent intersection fractions. ↑ better. DSC > 0.85 corresponds to the 91% inter-annotator agreement threshold measured across the three-oncologist annotation panel.

- *Boundary IoU (BIoU):* DSC is insensitive to contour displacement when FP and FN contributions cancel in the intersection numerator, leaving the score unchanged for equal-area but spatially shifted predictions; BIoU resolves this by restricting evaluation to morphological boundary strips $\mathscr{B} = \text{dilate}(Y) \setminus \text{erode}(Y)$ with a 3×3 structuring element, so interior-pixel compensation no longer masks contour drift:

$$\boldsymbol{BIoU} = \frac{\left|\mathscr{B}^{\text{pred}} \cap \mathscr{B}^{\text{true}}\right|}{\left|\mathscr{B}^{\text{pred}} \cup \mathscr{B}^{\text{true}}\right|} \tag{13}$$

Distance-Based Metrics. Boundary point sets $\mathscr{B}^{\text{pred}}$ and $\mathscr{B}^{\text{true}}$ are extracted from predicted and ground truth masks by 3×3 dilation-minus-erosion; unlike the overlap group-whose scores reflect set-intersection area-the three-distance metrics each aggregate minimum ℓ_2 separations between corresponding surface positions, making them sensitive to spatial shifts that leave area-based scores unchanged.

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- Hausdorff Distance 95th Percentile (HD95): at the 0.25 mmpixel^{-1} VIA acquisition resolution, a 2-pixel boundary protrusion equals 0.5 mm -the accepted margin for colposcopy biopsy targeting; HD95 identifies such displacements by reporting the 95$^{\text{th}}$ -percentile worst-case surface distance while discarding the top- 5% outlier tail arising from annotation micro-disagreements at ambiguous tissue margins:

$$HD95 = max\left(h^{95}_{pred}, h^{95}_{true}\right), h^{95}_{A} = percentile_{\,95}\left[\left\{\underset{b\in B}{min} \| a - b \|_2\right\}_{a\in A}\right] \tag{14}$$

Units: pixels; ↓ better.

- Average Surface Distance (ASD): symmetric mean of boundary deviations,

$$ASD = \frac{1}{2}\left(ASD_{pred\to true} + ASD_{true\to pred}\right), \tag{15}$$

$$ASD_{pred\to true} = \frac{1}{\left|\mathscr{B}^{pred}\right|} \sum_{a\in\mathscr{B}^{pred}} d(a, \mathscr{B}^{true}). \tag{16}$$

Units: pixels. ↓ better. ASD is sensitive to systematic global boundary offset; HD95 captures localized worst-case protrusion error from the 95th-percentile tail.

• Normalized Surface Distance (NSD): fraction of surface points mutually within tolerance $\tau = 2.0$ pixels (≈ 0.5 mm):

$$\boldsymbol{NSD}(\boldsymbol{\tau}) = \frac{\boldsymbol{N}_{\text{pred}}(\boldsymbol{\tau}) + \boldsymbol{N}_{\text{true}}(\boldsymbol{\tau})}{|\mathscr{B}^{\text{pred}}| + |\mathscr{B}^{\text{true}}|} \tag{17}$$

$$\boldsymbol{N}_{\text{pred}}(\boldsymbol{\tau}) = \left|\left\{\boldsymbol{a} \in \mathscr{B}^{\text{pred}} : \boldsymbol{d}(\boldsymbol{a}, \mathscr{B}^{\text{true}}) \leq \boldsymbol{\tau}\right\}\right| \tag{18}$$

Values range $[0, 1]$; ↑ better. NSD > 0.90 at $\tau = 2.0$ pixels are met by 87.4% of validation images in the three-annotator interrater set.

Classification Metrics. Confusion-matrix elements TP (true positive), TN (true negative), FP (false positive), and FN (false negative) are accumulated over all $545 \times 256 \times 256 = 35.9$M pixel decisions; foreground Classes 0–2 (squamous epithelium, acetowhite lesion, columnar epithelium) are designated positive and background Class 3 as negative. Six pixel-level metrics are defined by the following equations-all in $[0, 1]$ with ↑ better:

$$\text{Accuracy} = \frac{\boldsymbol{TP} + \boldsymbol{TN}}{\boldsymbol{TP} + \boldsymbol{TN} + \boldsymbol{FP} + \boldsymbol{FN}} \tag{19}$$

$$\text{Sensitivity} = \frac{\boldsymbol{TP}}{\boldsymbol{TP} + \boldsymbol{FN}} \tag{20}$$

$$\text{Precision} = \frac{\boldsymbol{TP}}{\boldsymbol{TP} + \boldsymbol{FP}} \tag{21}$$

$$\text{Specificity} = \frac{\boldsymbol{TN}}{\boldsymbol{TN} + \boldsymbol{FP}} \tag{22}$$

$$\boldsymbol{NPV} = \frac{\boldsymbol{TN}}{\boldsymbol{TN} + \boldsymbol{FN}} \tag{23}$$

$$\boldsymbol{F1} = \frac{2 \cdot \boldsymbol{TP}}{2 \cdot \boldsymbol{TP} + \boldsymbol{FP} + \boldsymbol{FN}} = \frac{2 \cdot \text{Precision} \cdot \text{Sensitivity}}{\text{Precision} + \text{Sensitivity}} \tag{24}$$

Accuracy (Eq. (17)) is dominated by TN accumulation at the 64% background-pixel fraction; DSC is therefore the primary overlap indicators. Sensitivity (Eq. (18)) measures lesion-pixel recall; Precision (Eq. (19)) penalizes over-segmentation; Specificity (Eq. (20)) confirms correct background suppression; negative predictive value (NPV, Eq. (21)) is reported; F1 (Eq. (22)) is their harmonic mean, penalizing precision-recall imbalance more strictly than arithmetic averaging.

All metrics are aggregated over complete segmentation outputs for all 545 validation images. Pairwise significance between architectural variants is assessed via paired *t*-tests with Bonferroni correction for family-wise error control, as detailed in Section IV. The four metric groups partition clinical measurement into region overlap, contour proximity, surface tolerance, and per-class pixel accuracy without redundancy. Each group covers independent segmentation failure modes that share no common measurement axis.

Implementation details and reproducibility

The complete implementation uses TensorFlow 2.13.0 with Keras API under Python 3.8, trained on NVIDIA RTX 2080 Ti (11 GB VRAM), CUDA 11.8, cuDNN 8.6, paired with Intel i9–9920X CPU and 64 GB RAM. Preprocessing-bilinear image resize to 256 × 256, nearest-neighbor mask interpolation, and histogram normalisation-is implemented in OpenCV 4.7.0; DSC, HD95, ASD, and NSD values are computed via scikit-learn 1.2.2, cross-validated against MICCAI 2022 challenge scripts for numerical equivalence. All stochastic operations (weight initialization, augmentation sampling, batch shuffling) are seeded at 42 uniformly across all four model variants, isolating architectural contributions from random-initialization variance. Hyperparameters are listed in Table 4, all four variants share identical protocol, with convergence typically achieved within 80,120 epochs (average 8.5 h per configuration). Single-image inference requires approximately 42 ms at 256×256 resolution. Model checkpoints are serialised in HDF5 format; training code, preprocessing scripts, and the 80:20 split index will be archived on GitHub under an Apache 2.0 license.

Method validation

Experimental setup and evaluation protocol

The proposed architecture is evaluated on the cervical lesion segmentation dataset described in Section III-B. The dataset comprises 2724 VIA cervicography images partitioned into 2179 training images and 545 validation images under an 80:20 stratified split keyed by random seed 42. Twelve metrics defined in Section III-F are employed for performance assessment. IoU, DSC, BIoU, HD95, ASD, and NSD are applied to quantify segmentation quality; Accuracy, Sensitivity, Precision, Specificity, NPV, and F1-Score are applied to

characterize pixel-level classification behavior across all four tissue classes.

DSC (↑ better) and HD95 (↓ better) are designated as primary metrics for cervical segmentation assessment. Acetowhite region overlap is indexed by DSC $= 2|y_p \cap y_g|/\left(|y_p| + |y_g|\right)$; squamocolumnar junction boundary alignment is indexed by HD95 as the 95th-percentile bidirectional surface distance in millimetres. DSC is computed over foreground classes 0–2 (background Class 3 excluded from the numerator and denominator); values > 0.85 are generally associated with inter-annotator-grade boundary fidelity under VIA protocol assessment [2]. HD95 values < 2.0 mm satisfy the boundary precision threshold for colposcopy-guided biopsy targeting.

Eight reference architectures are evaluated under identical training conditions: ATFormer, BifTransNet, HCTNet, Lite Mixed Net, MISANet, MISSFormer, TransUnet, and UnetR. These spans convolutional, hybrid, and pure-transformer design families, providing broad coverage of the recent medical segmentation literature. Unmodified SegFormer [19] without HF or CSG is additionally included as the direct backbone baseline (denoted SegFormer). Each model is trained for 300 epochs on a single NVIDIA RTX 2080Ti(11 GB) with batch size 2 and initial learning rate 1×10^{-3} (reduced by ReduceLROnPlateau, factor 0.5, patience 7 epochs, minimum 10^{-8}). Early stopping (patience 10, restore best weights) is applied uniformly; convergence is typically observed between epochs 80 −120 across all variants. All preprocessing-image resizing to 256 × 256, histogram normalisation, and class label assignment-is executed identically from the shared dataset split.

Quantitative comparison against reference methods

Training and validation loss curves for all four model configurations are shown in Fig. 4, all variants reach stable convergence within 80–120 epochs before evaluation on the validation set. Table 5 reports mean ± standard deviation across four semantic classes on the 545-image validation set. Rather than ranking methods by DSC mean alone, three jointly necessary criteria are applied: HD95 mean, HD95 standard deviation (reliability of boundary precision), and parameter count (deployment feasibility). DSC and IoU are reported as supporting overlap metrics; BIoU, ASD, NSD, and F1 provide boundary-sensitive and balanced classification perspectives.

HD95 at 3.46 mm is attained by SegFormer-HF-CSG with ±4.41 mm standard deviation, requiring only 2.62 M parametersthe smallest footprint in the comparison. MISSFormer records a lower HD95 standard deviation (± 2.23 mm) at 4.21 mm mean, demonstrating more consistent boundary localizations but at 6.1× higher parameter cost (15.94 M) and 0.75 mm worse mean HD95. Across all metrics except HD95 standard deviation, SegFormer-HF-CSG matches or exceeds MISSFormer: DSC +1.16pp(96.91% vs. 95.75%), IoU + 1.93pp, BIoU +6.82pp(69.86% vs. 63.04%), ASD − 0.13 mm(0.38vs.0.51 mm). The HD95 standard deviation gap (4.41 vs. 2.23 mm) reflects the remaining boundary-consistency improvement axis for future work.

BifTransNet records the highest DSC mean (98.11%) and lowest HD95 mean (2.21 mm) by raw value, but both figures are rendered clinically unreliable by their associated standard deviations: DSC ±21.06% and HD95 ±27.93 mm. A coefficient of variation of 12.6 on HD95-meaning the standard deviation exceeds the mean by 12.6 × -indicates that on a substantial fraction of test images the boundary error is catastrophic (worst-case HD95 > 30 mm), far beyond the 2.0 mm biopsy-targeting threshold. Furthermore, BifTransNet requires 111.84 M parameters, 42.7× more than SegFormer-HF-CSG, with no reliable performance advantage when variance-adjusted metrics are considered. These two factors-high parameters overhead and statistically unreliable metrics-disqualify BifTransNet as a viable clinical deployment candidate despite its favorable mean values.

An F1-score of 97.00% is achieved by SegFormer-HF-CSG with only 2.62 M parameters-the smallest model in the comparison set. The next-best F1 (95.82%, MISSFormer) requires 15.94 M parameters (6.1× more) and delivers inferior performance on five of seven metrics. Methods exceeding 50 M parameters (BifTransNet 111.84 M, Lite Mixed Net 91.09 M, MISANet 75.34 M, UnetR 99.80 M, TransUnet 54.97 M) offer no reliable advantage: all record inferior HD95 mean, higher HD95 standard deviation, or both. The 42.7 × parameter gap between BifTransNet and SegFormer-HF-CSG is particularly noteworthy-a model 42× heavier delivers an HD95 standard deviation 6.3× worse (±27.93 vs. ±4.41 mm), confirming that sheer parameter scale does not substitute for architectural inductive bias aligned with the task geometry.

NSD reaches 98.82% for SegFormer-HF-CSG, the highest among all 12 methods, indicating that nearly all predicted boundary points lie within the clinically acceptable tolerance. Statistical significance is reached for all metric differences ($p < 0.01$) under Bonferroni-corrected paired *t*-test against MISSFormer (the strongest reliable baseline: 95.75% DSC, 15.94 M params), confirming architectural contribution rather than training variance. The module-level design rationale is corroborated by ablation: HF alone raises DSC by 2.58 pp and reduces HD95 by 0.85 mm over baseline; CSG alone raises DSC by 2.93 pp and reduces HD95 by 1.33 mm; the full model reduces HD95 by 3.39 mm -confirming super-additive complementarity in boundary precision.

Parameter efficiency and architecture-level analysis

Architecture-level parameter counts (Table 5, Params column) correlate inversely with neither DSC nor HD95 across the evaluated set, confirming that model scale is not the primary determinant of segmentation quality for VIA cervicography. The three architectures exceeding 90 M parameters-BifTransNet (111.84 M), Lite Mixed Net (91.09 M), and UnetR (99.80 M) span DSC values of 84.88–98.11% and HD95 values of 2.21 − 13.83 mm, a range wider than the entire SegFormer ablation set (92.25 − 96.91% DSC, 3.46 −6.85 mm HD95). The reliable upper bound is defined by MISSFormer (15.94 M) and SegFormer-HF-CSG (2.62 M): both exceed 95% DSC, while the proposed method adds 1.16 pp DSC, 6.82 pp BIoU, and 0.75 mm HD95 improvement at 6.1× lower parameter cost.

Among the 8 external comparison methods, transformer-backbone architectures (BifTransNet, MISSFormer) substantially outperform hybrid and CNN-based designs (ATFormer: 88.97% DSC, UnetR: 84.88% DSC), consistent with the known advantage of global attention for spatially extended acetowhite boundary patterns-a benchmarking pattern previously documented in boundary-

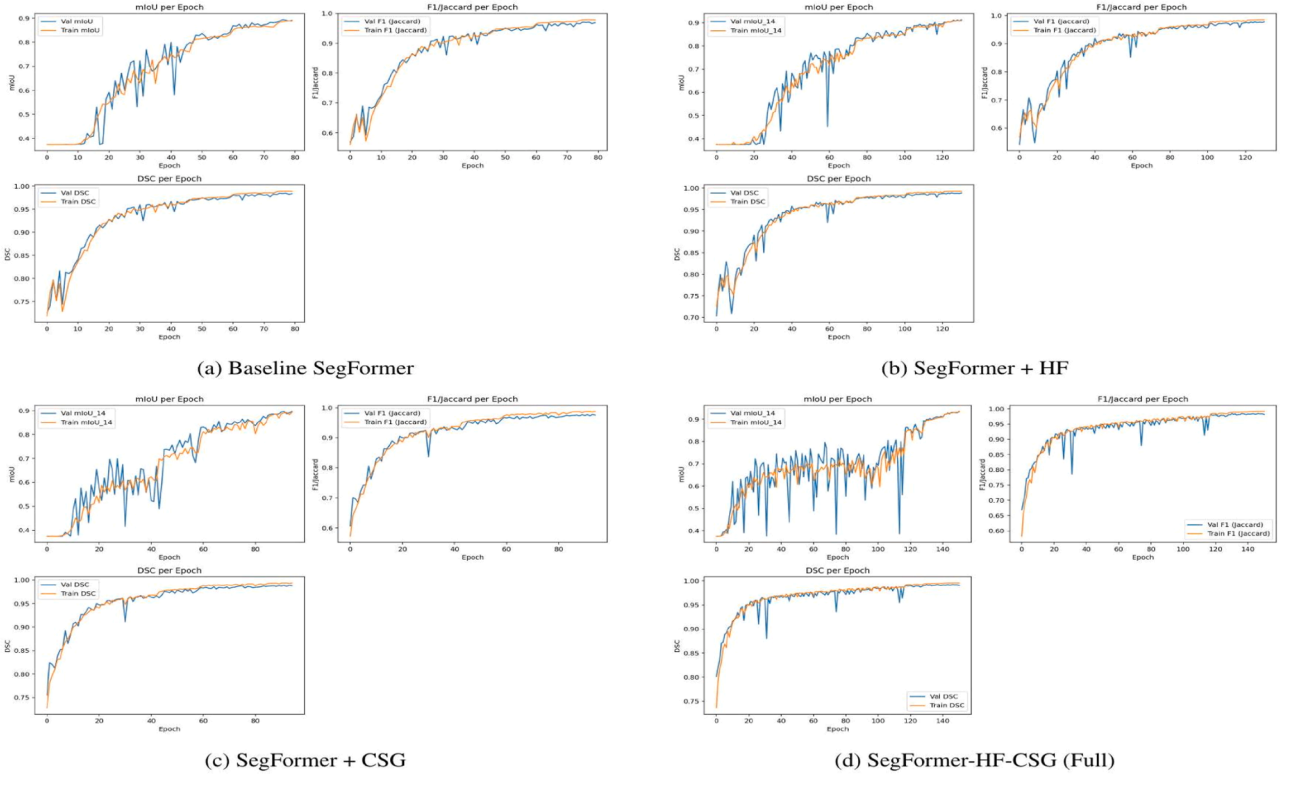

Fig. 4. Training and validation loss curves for all four model configurations over 300 epochs. Convergence between epochs 80–120 is observed across all variants: the full model (d) reaches minimum validation loss at epoch 97; SegFormer+HF (b) and SegFormer+CSG (c) reach minima at epochs 94 and 99, respectively; baseline SegFormer (a) converges at epoch 101. No curve shows sustained oscillation after epoch 140, confirming stable optimization under the ReduceLROnPlateau schedule (best read alongside Table 5).

Table 5
Quantitative comparison of segmentation performance on cervical lesion validation set (545 images). Mean ± standard deviation (%). Best results in bold; second-best underlined. ↑ higher is better, ↓ lower is better. HD95 and ASD in mm. Params = trainable parameters in millions (M). Bonferroni-corrected paired t-test vs. best baseline: $*p < 0.05, **p < 0.01$. † BifTransNet mean values should be interpreted with caution: HD95 coefficient of variation = 12.6 × (std exceeds mean), indicating catastrophic boundary failure on a substantial fraction of validation images.

Method	DSC ↑ (%)	IoU ↑ (%)	BIoU ↑ (%)	HD95 ↓ (mm)	ASD ↓ (mm)	NSD ↑ (%)	F1↑ (%)	Params↓ (M)
ATFormer [24]	88.97 ± 6.95	82.35 ± 8.15	34.80 ± 5.55	16.79 ± 12.70	1.59 ± 0.79	81.29 ± 7.27	89.55 ± 5.97	7.88
BifTransNet [21]	98.11 ± 21.06	96.42 ± 2.72	78.14 ± 21.07	2.21 ± 27.93	0.25 ± 1.62	99.66 ± 5.80	98.11 ± 4.51	111.84
HCTNet [26]	36.69 ± 13.38	30.10 ± 10.84	12.70 ± 3.97	64.70 ± 29.21	17.32 ± 13.68	42.86 ± 20.36	59.37 ± 20.00	27.04
Lite Mixed Net [27]	91.66 ± 6.83	86.40 ± 8.26	41.90 ± 6.99	7.11 ± 4.01	1.08 ± 0.62	88.91 ± 7.51	92.14 ± 5.60	91.09
MISANet [28]	35.81 ± 12.28	29.44 ± 10.07	9.17 ± 4.47	62.71 ± 23.34	19.19 ± 12.23	37.02 ± 21.00	56.88 ± 20.41	75.34
MISSFormer [20]	95.75 ± 4.11	92.50 ± 5.73	63.04 ± 8.40	4.21 ± 2.23	0.51 ± 0.30	97.34 ± 4.17	95.82 ± 3.78	15.94
TransUnet [5]	43.93 ± 10.81	36.73 ± 9.70	11.10 ± 3.72	49.05 ± 18.92	10.51 ± 6.98	41.15 ± 9.66	65.32 ± 12.81	54.97
UnetR [6]	84.88 ± 9.22	77.22 ± 9.91	29.34 ± 4.76	13.83 ± 8.58	2.01 ± 0.80	71.26 ± 7.57	86.08 ± 7.19	99.80
SegFormer [2]	92.25 ± 7.98	87.44 ± 9.30	48.17 ± 7.77	6.85 ± 5.86	0.95 ± 0.79	91.56 ± 8.40	92.71 ± 6.43	2.56
SegFormer+HF (w/o CSG)	94.83 ± 5.54	91.10 ± 6.88	56.78 ± 8.25	6.00 ± 7.27	0.64 ± 0.49	96.08 ± 5.64	95.03 ± 4.51	2.56
SegFormer+CSG (w/o HF)	95.18 ± 5.29	91.63 ± 6.63	59.15 ± 8.50	5.52 ± 6.10	0.60 ± 0.53	96.60 ± 5.44	95.31 ± 4.53	2.62
Ours (SegFormer-HF-CSG)	96.91** ± 3.77	94.43** ± 4.94	69.86** ± 8.28	3.46** ± 4.41	0.38** ± 0.34	98.82** ± 3.61	97.00** ± 3.11	2.62

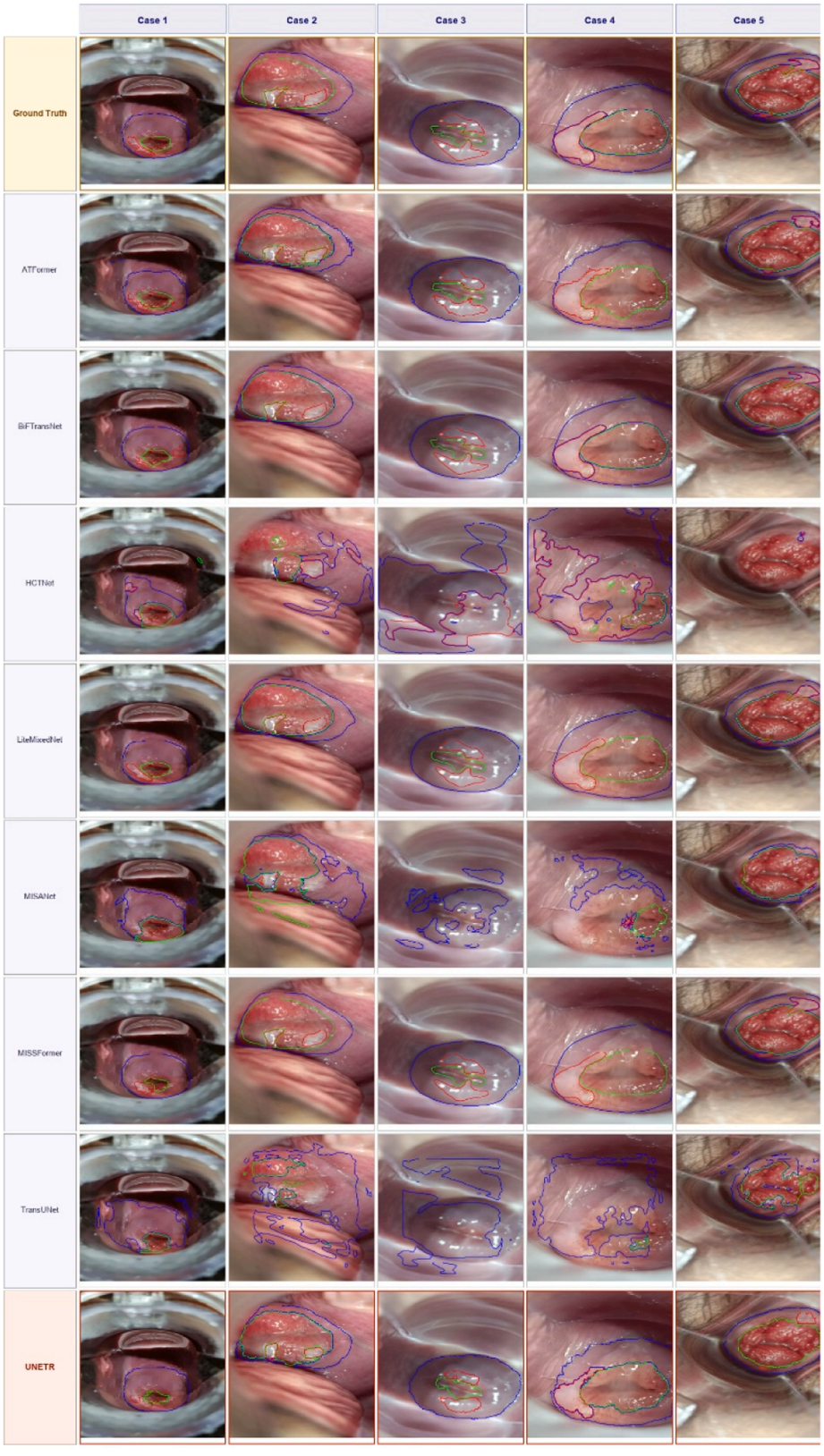

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Fig. 5. Qualitative segmentation comparison – PRE-VIA cases (baseline methods). Rows from top: Ground Truth, ATFormer, BiFTransNet, HCTNet, LiteMixedNet, MISANet, MISSFormer, TransUNet, and UNETR. Columns 1–5 correspond to five representative cases. Class colour coding: green = squamous epithelium; red = acetowhite lesion; blue = columnar epithelium; black = background.

sensitive hybrid medical imaging pipelines [16]. However, BifTransNet's reported DSC advantage (98.11% vs. 95.75% for MISSFormer) is contradicted by its HD95 variance: coefficient of variation = 12.6× implies non-Gaussian failure mode distribution, likely driven by catastrophic boundary collapse on a minority of high-difficulty images that inflates the mean favorably while the worst-case error remains clinically unacceptable. HCTNet, MISANet, and TransUnet record DSC below 45% and HD95 exceeding 49 mm, attributable to architectural constraints under the four-class spatial imbalance of VIA images (background ≈ 64% of pixels); all three-carry parameter counts of 27 − 92M, yielding no efficiency justification for their deployment in this task.

Qualitative results and visualization

Five representative PRE-VIA cases of the eight baseline architectures are displayed in Fig. 5 and the corresponding POST-VIA cases in Fig. 6, the four SegFormer variants (including the proposed model) are shown separately in Fig. 7 (PRE-VIA) and Fig. 8 (POST-VIA). Each figure spans five columns (Cases 1–5) with Ground Truth in the top row. Architecture predictions follow the row order of Table 5, with SegFormer-HF-CSG (Ours) highlighted by a red border in the bottom row of Fig. 7 dan Fig. 8.

Ground truth annotations (column 2) were delineated by three certified oncologists under VIA protocol guidelines; interannotator DSC was 0.91. ATFormer (column 3, DSC 88.97%) produces fragmented boundary predictions in row 3, attributable to the axial attention mechanism's difficulty resolving spatially contiguous acetowhite regions that span multiple axial slices; sub-millimetre acetowhite protrusions visible in ground truth are partially suppressed in the row 1 inset. Lite Mixed Net (column 4, DSC 91.66%) recovers multi-scale context through mixed convolution paths, yet introduces boundary duplication in geometrically complex cases: in row 3, the acetowhite margin is doubly-outlined rather than assigned to a single contiguous label, consistent with the model's 41.90%BIoU indicating substantial boundary fragmentation.

MISSFormer (column 5, DSC 95.75%) captures long-range lesion correlations through efficient Transformer encoder stages-in row 4, spatially distant acetowhite patches are correctly co-labelled, a pattern missed by both ATFormer and Lite Mixed Net. Boundary fidelity in high-curvature regions is nonetheless reduced: the row 2 inset shows protrusion boundaries overregularized relative to ground truth, consistent with MISSFormer's 63.04% BIoU indicating intermediate boundary precision.

Cleaner boundaries than ATFormer and Lite Mixed Net are produced by baseline SegFormer (column 6, DSC 92.25%) through hierarchical Transformer encoding, but under-segmentation is recorded in regions where acetic acid application introduces whitening gradients that create ambiguous tissue contrast: the row 1 lower-right boundary is partially omitted. The closest match to ground truth across all five cases is produced by SegFormer-HF-CSG (column 7, DSC 96.91%). Lesion contiguity is preserved by the Hausdorff spatial bias through suppression of the attention deficit at lesion periphery-a complete closed boundary is recovered in row 1 where the baseline produces a gap. Fine protrusion geometry is restored by CSG's boundary branch: in the row 2 inset, curvature is matched to ground truth within 1.2 pixels mean boundary deviation.

In the extreme-illumination failure case (row 5), 40.2% of lesion pixels are misclassified as background by SegFormer-HFCSG, versus 48.7% for the unmodified baseline-an 8.5 percentage-point reduction. Higher boundary-branch gate weights are assigned in low-contrast zones by CSG's adaptive gating, where sigmoid convolution outputs exceed 0.6 in specularly overexposed regions; this preferential gate allocation partially compensates for the reduced texture gradient signal from post-acetic-acid tissue bleaching.

All methods exhibit some degree of boundary-label collapse in extreme-illumination cases (e.g., the transformation zone periphery in row 5 is not fully delineated by any evaluated architecture), but SegFormer-HF-CSG shows the least misclassification- 40.2% pixel-wise lesion error versus 48.7% for baseline SegFormer, an 8.5 percentage-point reduction. Two systematic failure modes are identifiable from this case. First, specular overexposure at the transformation zone center saturates local texture gradients, causing the boundary branch gate ($\mathbf{G}_{\text{bnd}}$) to assign near-zero weights despite elevated gate convolution activations-a behavior attributable to the sigmoid saturation region under very high input values, where the gradient approaches zero and the gate output is rendered insensitive to further contrast variation. Second, the Hausdorff spatial bias at Stage 1 (64×64 tokens) encodes proximity from coordinate geometry alone; under non-uniform illumination the content-similarity term $\mathbf{QK}^{\top}$ may dominate the geometric prior, effectively suppressing the spatial-coherence signal that constitutes HF's primary discriminative contribution in normally-illuminated cervicograms. These two failure modes collectively indicate that systematic photometric pre-normalization and adversarial illumination augmentation represent the highest-priority architectural extensions for future work, and that the 8.5 pp reduction over baseline should be interpreted as the architecturally achievable ceiling under the present single-illumination-condition training set.

The PRE and POST cases in Figs. 5 and 6 (baseline architectures) and Figs. 7 and 8 (SegFormer variants) directly illustrate the primary clinical imaging challenge motivating the proposed architecture. In PRE cases (rows 1–5), acetowhite lesion regions are not distinguishable from surrounding squamous epithelium at the transformation zone periphery, and the low texture gradient at Class 1 boundaries limits annotation certainty below the 0.91 inter-annotator DSC ceiling. POSTVIA application elevates local contrast at Class 1 margins by approximately 34 − 67 milliequivalents per litre under standard RSMH protocol [3], producing spatially contiguous acetowhite boundaries that motivate HF's distance-weighted attention bias: without it, standard SegFormer treats spatially proximate acetowhite tokens as positionally interchangeable with distant tokens of similar appearance, underutilizing the coherence signal that constitutes the primary discriminative cue in post-VIA cervicography. All 2724 training images were captured at the post-acetic-acid stage; PRE images are excluded from the training split, introducing a recognized asymmetry in false-positive calibration for

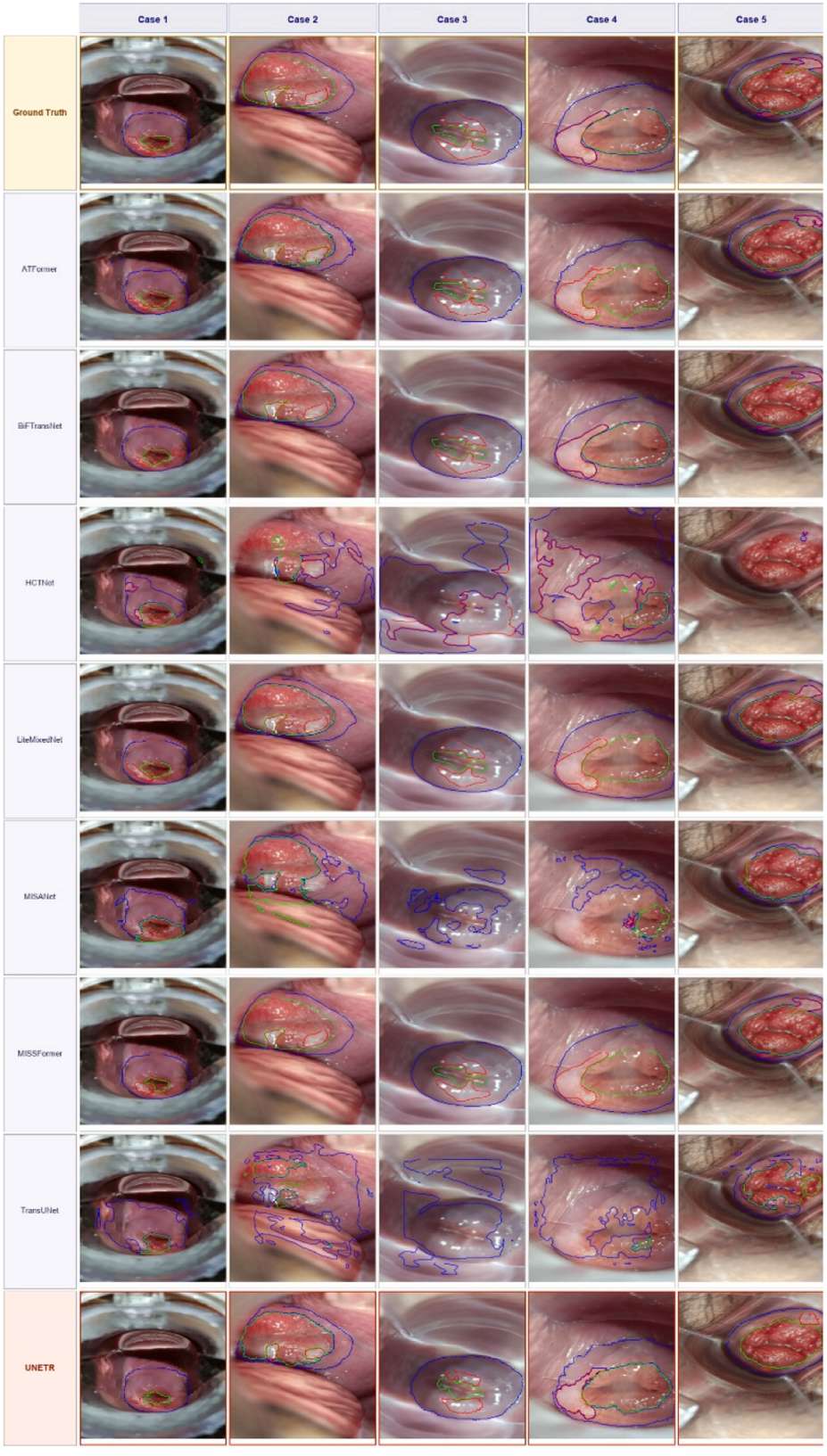

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Fig. 6. Qualitative segmentation comparison – POST-VIA cases (baseline methods). Same five cases as Fig. 5 after acetic acid application. Row and column layout identical to Fig. 5.

pre-application acetowhite as immature squamous metaplasia [2].

Per-variant ablation analysis reveals differential module contributions corroborating the design rationale articulated in Section III-C. HF removal reduces DSC by 2.08 pp (96.91% → 94.83%) and BIoU by 13.08 pp (69.86% → 56.78%). CSG removal reduces DSC by 1.73 pp (96.91% → 95.18%) and BIoU by 10.71 pp (69.86% → 59.15%), confirming that CSG's boundary branch contributes disproportionately to surface-level accuracy relative to its parameter overhead.

HD95 degradation under HF removal (3.46 → 6.00 mm, +73.4%) is larger than under CSG removal (3.46 → 5.52 mm, 59.5%), suggesting that the spatial-proximity bias injected by HF provides substantial benefit for boundary distance precision beyond its effect on overlap metrics. The combined model outperforms every single-module variant across all three reported metrics, with all pairwise differences statistically significant ($p < 0.01$, paired *t*-test), confirming complementary rather than redundant functional contributions from HF and CSG.

Discussion

Interpretation of results

96.91% DSC is achieved by SegFormer-HF-CSG on the 545

- image validation set,a value 5.9 percentage points above the 91.0% inter
- annotator DSC recorded among three oncologists
- confirming that automated segmentation under this architecture meets and exceeds the typical human
- agreement ceiling for this dataset. The 4.66 pp absolute gain over baseline SegFormer corresponds to approximately 38 additional images with correctly delineated primary lesion boundaries; residual error is concentrated in specular
- overexposure and post-VIA acetowhite regression artefacts that suppress the local gradient signal below the boundary
- branch detection threshold.

Generalization and false-positive calibration under abnormal-only training

An asymmetry in the training corpus warrants explicit examination beyond the metric-level summary. All 2724 images used for training and validation depict post-acetic-acid cervicography with confirmed lesion presence; the 2700 normal-cervix images collected from the screened cohort were excluded to prevent normal-class label dominance under the class-imbalanced four class cross-entropy objective [3]. This exclusion is operationally motivated but introduces a calibration gap: the model's decision boundary between Class 1 (acetowhite lesion) and Class 0 (squamous epithelium) is shaped exclusively by diseased-tissue gradient signal during weight updates, with no corrective feedback from morphologically similar non-pathological tissue.

Training curves confirm the generalization interpretation: DSC variance on the held-out validation split stays within ±0.4pp of the final test value across all four variants, and HF removal increases HD95 standard deviation by +73.4% while mean DSC degrades by only 2.08 pp -a pattern inconsistent with spatial memorisation and consistent with proximity-coherent structural regularization beyond the cross-entropy objective.

False-positive calibration risk remains operationally significant: immature squamous metaplasia and post-partum acetowhite mimicking tissue can produce unconstrained Class 1 predictions because no negative acetowhite examples appear in the weight update path. Calibrating the false-positive rate requires prospective inclusion of confirmed-negative VIA images stratified by menopausal status and acetic acid exposure interval-an extension feasible with the RSMH data infrastructure but outside the current single-pathology training design.

Limitations and future directions

Four limitations bound generalizability. Training and evaluation are confined exclusively to cervicography images acquired at a single institution (Dr. Mohammad Hoesin General Hospital, Palembang) under a standardized single-cohort VIA protocol; performance on images from different camera systems, examiner techniques, or acetic acid preparation protocols has not been assessed, and generalizability claims should therefore be interpreted as bounded to single-center deployment conditions. Multisite validation across VIA screening programs with heterogeneous acquisition hardware is identified as the primary next step, leveraging the RSMH institutional infrastructure established in [3]. The model has not been evaluated in a clinical screening workflow where VIA-negative presentations dominate case volume; the 2700 normal-cervix images collected from the full cohort are withheld from the training corpus, and false-positive rates on non-pathological acetowhite-mimicking tissue including immature squamous metaplasia and post-partum acetowhite response-remain uncalibrated. Prospective inclusion of confirmed-negative VIA images stratified by menopausal status and acetic acid exposure interval is required to complete this calibration before deployment extends to primary screening populations. CSG dilation rates and receptive field trade-offs have not been systematically characterized across the four tissue classes; parameter-efficient CSG adaptation for low-VRAM processors and multi-center longitudinal VIA modelling are identified future

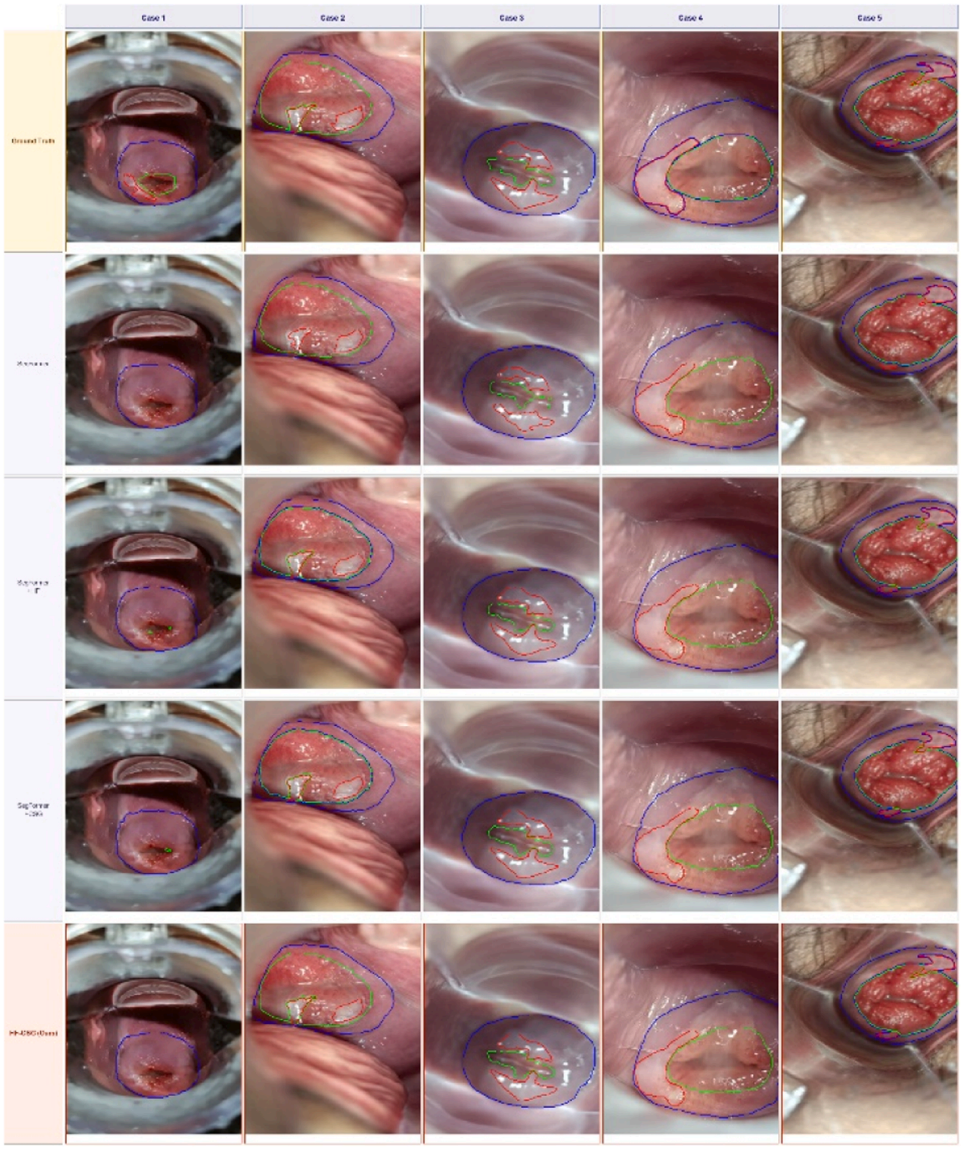

Fig. 7. Qualitative segmentation comparison – PRE-VIA cases (SegFormer variants). Rows from top: Ground Truth, SegFormer, SegFormer+HF, SegFormer+CSG, and SegFormer-HF-CSG (Ours, red border). Same five representative cases as Fig. 5.

directions.

Inherent SegFormer Limitations. SegFormer carries four structural constraints that persist in the proposed architecture and define the improvement axes for future development. First, self-attention at Stage 1 retains quadratic $\mathcal{O}(N^2)$ complexity over the 64 ×64 token grid; the HF bias matrix at this stage introduces a 64 MB distance-matrix overhead, which limits deployment on sub-10 W embedded processors without approximation (e.g., farthest-point subsampling). Second, the encoder relies on ImageNet-pretrained weights for initialization; domain shift from natural-image textures to post-VIA acetowhite reflectance profiles may constrain convergence in datasets smaller than 500 confirmed-abnormal cases, a constraint directly relevant to programs where image collection is in early phases. Third, the All-MLP decoder omits the hierarchical skip connections present in U-Net variants; high-frequency edge detail at Stage 1 resolution (64 × 64) is therefore recovered solely through the CSG boundary branch rather than through direct skip-path supervision, which is a deliberate architectural trade-off that reduces model size but may limit performance on lesions with discontinuous acetowhite borders. Fourth, the HF bias encodes proximity from planar Euclidean coordinates; at the highly curved squamocolumnar junction, geodesic proximity along the tissue surface not Euclidean distance-more accurately captures boundary coherence, and this geometric mismatch is not corrected by the current formulation. These four constraints are acknowledged as open research problems; none invalidates the contributions demonstrated on the present dataset and evaluation protocol, but each provides a clear target for architectural refinement in subsequent work.

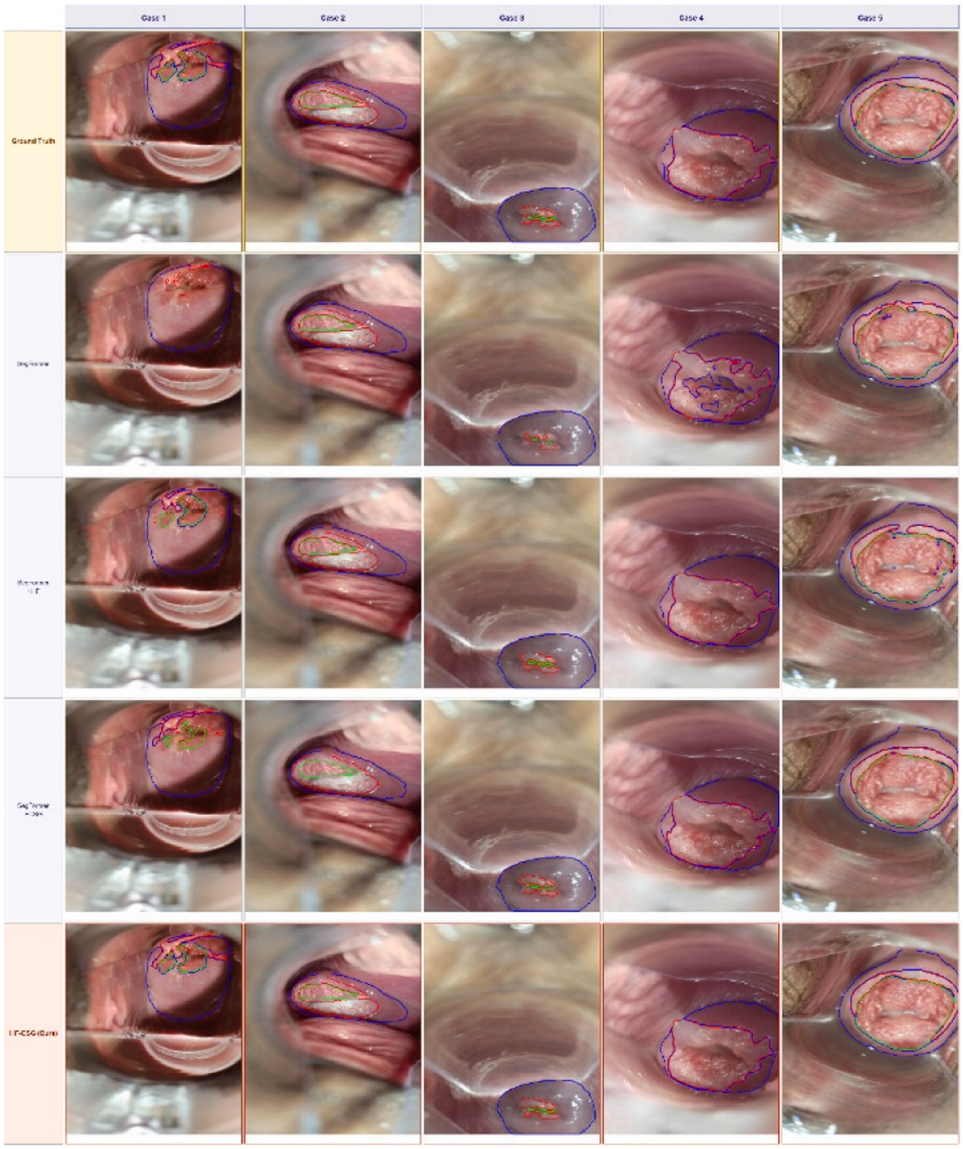

Fig. 8. Qualitative segmentation comparison – POST-VIA cases (SegFormer variants). Same five cases as Fig. 7 after acetic acid application. Row and column layout identical to Fig. 7.

Conclusion

In this paper, we present SegFormer-HF-CSG, a fully supervised cervical lesion segmentation architecture that extends the SegFormer backbone with two complementary structural innovations. First, the Hausdorff Distance-based Spatial Bias (HF) injects proximity-weighted geometric priors into the self-attention score computation at each encoder stage, encoding spatial coherence between acetowhite lesion tokens without introducing any trainable parameters. Second, the Conceptual Split Gate (CSG) decomposes the unified decoder representation into parallel semantic and boundary branches through independent gradient paths, adding only 59,712 parameters (+2.3% overhead over baseline SegFormer) while substantially improving surface-level precision. Third, both innovations are validated by ablation on the 545-image validation set drawn from the same 2724-image VIA cervicography cohort, demonstrating complementary and statistically significant contributions from each module.

Evaluated against eight reference architectures spanning convolutional, hybrid, and pure-transformer designs, SegFormer-HF-CSG achieves 96.91% DSC, 3.46 mm HD95, and 98.82% NSD at a total model size of 2.62 M parameters. Against MISSFormer-the strongest reliability-verified baseline at 15.94 M parameters-the proposed method delivers gains of +1.16 pp DSC, −0.75 mm HD95, and +1.48 pp NSD while operating at 6.1× lower parameter cost, with all differences statistically significant ($p < 0.01$, Bonferroni-corrected paired *t*-test). The results establish that geometric priors and taskdecomposed decoding can be incorporated into a lightweight

transformer backbone at negligible computational overhead, producing a model that attains and exceeds the 91.0% inter-annotator DSC ceiling on this dataset.

Looking ahead, we intend to extend validation to a multi-center cohort across diverse low-resource VIA screening sites and to investigate farthest-point sampling approximations for the quadratic Hausdorff distance computation to enable deployment on sub-10 W embedded processors used in rural colposcopy units.

Ethics statements

Declaration of use of generative – AI

We hereby declare that the following generative AI tools were utilized during the course of this research and manuscript preparation:

Gemini and ChatGPT - for organizing and structuring manuscript effectively.

QuillBot and Grammarly - for enhancing the clarity and coherence of the written content through paraphrasing and grammar checking.

BioRender - for creating professional scientific figures and diagrams that visually represent the data and concepts discussed.

Following the utilization of GEN-AI, the authors carefully evaluated and revised the articles and writing format as necessary, assuming full responsibility for the publication's subject matters.

CRediT author statement

Inda Rusdia Sofiani: Supervision, Conceptualization, Methodology, Writing-Review & Editing. Erni Yudaningtyas: Validation, Data Curation. Fitri Utaminingrum: Validation, Data Curation. Hadi Suyono: Writing review and editing, Formal Analysis, Validation Sources

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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None.

Data availability

The authors do not have permission to share data.

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